Critical Analysis of CRPD Alternative Reports: A Computational Approach to Disability Rights Monitoring

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1 Introduction

This project examines how civil society organizations assess State Party implementation of the United Nations Convention on the Rights of Persons with Disabilities (CRPD) by applying computational text analysis to alternative (shadow) reports submitted to the CRPD Committee. While existing research and monitoring practices rely heavily on government self-reports, this study centers civil society perspectives to identify policy priorities, implementation gaps, and patterns that may be obscured in official reporting. Using a large corpus of alternative reports across regions and institutional contexts, the project demonstrates the value of triangulated, multi-informant approaches for evaluating disability inclusion and accountability under the CRPD.

2 Data Collection and Preparation

The data for this project will be collected from the United Nations Treaty Body Database, specifically the OHCHR portal for treaty body documentation, which provides public access to civil society alternative (shadow) reports submitted to the Committee on the Rights of Persons with Disabilities UN Treaty Body Database. This platform hosts alternative reports across countries, reporting cycles, and institutional contexts. Data collection will be conducted programmatically using Python with Selenium, enabling automated navigation of the database, systematic retrieval of report metadata, and downloading of report documents in multiple formats. The full data collection script is provided below to ensure transparency, reproducibility, and methodological rigor.

2.1

Reticulate setup

The belt and suspenders approach to ensure the use of correct Python environment. This along with the python path specified in the YAML header, and the Select Interpreter button will ensure the correct Python environment is used for this lab for both interactive work and when rendering the document.

```
library(reticulate)

# Set Python path BEFORE loading spacyr - this is critical!
Sys.setenv(RETICULATE_PYTHON =
 ↪  '/opt/anaconda3/envs/TextMiningLLM/bin/python')
Sys.setenv(SPACY_PYTHON = '/opt/anaconda3/envs/TextMiningLLM/bin/python')

use_condaenv("TextMiningLLM", required = TRUE)
```

2.2 CRPD Alternative reports data collection with Python

```python
#| label: web-scraping
import os
import time
import requests
from selenium import webdriver
from selenium.webdriver.chrome.service import Service
from selenium.webdriver.chrome.options import Options
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
from webdriver_manager.chrome import ChromeDriverManager
from urllib.parse import urljoin

# Configuration
BASE_URL =
 ↪  "https://tbinternet.ohchr.org/_layouts/15/treatybodyexternal/TBSearch.aspx?Lang=en"
DOWNLOAD_DIR = "downloads"
MAX_PAGES = None  # Set to None for all pages

def setup_driver():
    options = Options()
    options.add_argument("--headless")  # Comment out to see the browser
    # options.add_argument("--disable-gpu")
    options.add_argument("--no-sandbox")
    options.add_argument("--disable-dev-shm-usage")
    options.add_argument("--window-size=1920,1080")
    # options.add_argument("--remote-debugging-port=9222") # Optional
     ↪  stability

    # Setup download directory in Chrome (just in case we need fallback)
    prefs = {
        "download.default_directory": os.path.abspath(DOWNLOAD_DIR),
        "download.prompt_for_download": False,
        "directory_upgrade": True
    }
    options.add_experimental_option("prefs", prefs)

    service = Service(ChromeDriverManager().install())
    driver = webdriver.Chrome(service=service, options=options)
    return driver
```

```python
def download_file(url, session, cookies, filename=None):
    try:
        response = session.get(url, cookies=cookies, stream=True)
        response.raise_for_status()

        # Check if it's actually a file
        content_type = response.headers.get('Content-Type', '')
        if 'text/html' in content_type:
            print(f"  [!] URL returned HTML instead of file: {url}")
            # Save the HTML for debugging
            try:
                with open(f"error_{int(time.time())}.html", "w") as f:
                    f.write(response.text)
            except: pass
            return False

        # Determine filename
        if not filename:
            if "Content-Disposition" in response.headers:
                import re
                fname = re.findall("filename=(.+)",
↪   response.headers["Content-Disposition"])
                if fname:
                    filename = fname[0].strip('"')

            if not filename:
                filename =
↪   url.split("symbolno=")[-1].split("&")[0].replace("/", "_") + ".pdf" #
↪   Default fallback

        filepath = os.path.join(DOWNLOAD_DIR, filename)

        # Avoid re-downloading
        if os.path.exists(filepath):
            print(f"  [i] File already exists: {filename}")
            return True

        with open(filepath, 'wb') as f:
            for chunk in response.iter_content(chunk_size=8192):
                f.write(chunk)
        print(f"  [+] Downloaded: {filename}")
        return True
```

```
    except Exception as e:
        print(f"  [!] Error downloading {url}: {e}")
        return False

def main():
    if not os.path.exists(DOWNLOAD_DIR):
        os.makedirs(DOWNLOAD_DIR)

    print("Setting up driver...")
    driver = setup_driver()
    print("Driver setup complete.")
    wait = WebDriverWait(driver, 20)

    # Configuration to resume
    START_PAGE = 1 # Start from beginning, file check will skip existing

    try:
        print("Navigating to search page...")
        driver.get(BASE_URL)

        # 1. Filter by Committee: CRPD
        print("Applying 'CRPD' filter...")
        try:
            # Strict text match to avoid matching CRPD-OP-AC etc.
            crpd_element =
↪   wait.until(EC.presence_of_element_located((By.XPATH,
↪   "//span[contains(@class, 'rlbText') and
↪   normalize-space(text())='CRPD']")))
            driver.execute_script("arguments[0].scrollIntoView(true);",
↪   crpd_element)

            crpd_label = crpd_element.find_element(By.XPATH,
↪   "./parent::label")
            crpd_checkbox = crpd_label.find_element(By.CLASS_NAME,
↪   "rlbCheck")

            if not crpd_checkbox.is_selected():
                print("Clicking 'CRPD' filter...")
                crpd_element.click()
                time.sleep(5) # Wait for AJAX update

            # Re-verify after potential DOM update
```

```
            try:
                crpd_element = driver.find_element(By.XPATH,
↪ "//span[contains(@class, 'rlbText') and normalize-space(text())='CRPD']")
                crpd_label = crpd_element.find_element(By.XPATH,
↪ "./parent::label")
                crpd_checkbox = crpd_label.find_element(By.CLASS_NAME,
↪ "rlbCheck")

                if not crpd_checkbox.is_selected():
                    print("Retry clicking 'CRPD' checkbox directly...")
                    driver.execute_script("arguments[0].click();",
↪ crpd_checkbox)
                    time.sleep(5)
            except:
                pass

            print("'CRPD' filter applied.")

        except Exception as e:
            print(f"Error applying CRPD filter: {e}")
            raise e

        # 2. Filter by Document Type Category: Reporting cycle related
        print("Applying 'Reporting cycle related' filter...")
        # Use contains to be safer
        try:
            cat_span = wait.until(EC.element_to_be_clickable((By.XPATH,
↪ "//span[contains(@class, 'rlbText') and contains(text(), 'Reporting cycle
↪ related')]")))
            cat_label = cat_span.find_element(By.XPATH, "./parent::label")
            cat_checkbox = cat_label.find_element(By.CLASS_NAME, "rlbCheck")

            if not cat_checkbox.is_selected():
                print("Clicking 'Reporting cycle related' span...")
                cat_span.click()
                time.sleep(2)

                # Re-find elements to avoid stale reference after click
                print("Re-verifying selection...")
                try:
                    cat_span = driver.find_element(By.XPATH,
↪ "//span[contains(@class, 'rlbText') and contains(text(), 'Reporting cycle
↪ related')]")
```

```
                    cat_label = cat_span.find_element(By.XPATH,
↪  "./parent::label")
                    cat_checkbox = cat_label.find_element(By.CLASS_NAME,
↪  "rlbCheck")

                    # Double check
                    if not cat_checkbox.is_selected():
                        print("Checkbox not selected after span click.
                         ↪  Clicking checkbox directly...")
                        driver.execute_script("arguments[0].click();",
↪  cat_checkbox)
                        time.sleep(2)
                except Exception as e:
                    print(f"Error re-verifying/clicking: {e}")
            else:
                print("'Reporting cycle related' already selected.")

            print("Waiting for AJAX update (10s)...")
            time.sleep(10) # Give it plenty of time for the Document Type
↪  list to repopulate
        except Exception as e:
            print(f"Skipping Document Type Category filter (User requested
             ↪  all types): {e}")

        # 3. Filter by Document Type: Info from Civil Society Organizations
         ↪  (for LOIs)
        print("Applying 'Info from Civil Society Organizations (for LOIs)'
         ↪  filter...")

        print("Waiting for '(for LOIs)' option to appear...")
        try:
            # Wait up to 20 seconds for the text to appear in the specific
             ↪  container
            wait.until(lambda d: "(for LOIs)" in d.find_element(By.ID,
↪  "ctl00_ContentPlaceHolder1_rlbDocType").text)
            print("Found '(for LOIs)' text in the list!")
        except:
            print("Timed out waiting for '(for LOIs)' text. Dumping current
             ↪  list content for debugging...")
            try:
                list_el = driver.find_element(By.ID,
↪  "ctl00_ContentPlaceHolder1_rlbDocType")
```

```
                print(f"List content preview: {list_el.text[:500]}...")
            except: pass

        # Get all items again
        items = driver.find_elements(By.XPATH,
↪  "//div[@id='ctl00_ContentPlaceHolder1_rlbDocType']//span[contains(@class,'rlbText')]")

        target_found = None

        for item in items:
            text = item.text.strip()
            # Strict check based on screenshot
            if "(for LOIs)" in text and "Civil Society" in text:
                target_found = item
                break

        if target_found:
            print(f"Selecting filter: {target_found.text}")
            driver.execute_script("arguments[0].scrollIntoView(true);",
↪  target_found)

            label = target_found.find_element(By.XPATH, "./parent::label")
            checkbox = label.find_element(By.CLASS_NAME, "rlbCheck")

            if not checkbox.is_selected():
                target_found.click()
                time.sleep(2)
        else:
            print("Specific '(for LOIs)' filter not found. Trying to find any
             ↪  'Civil Society' option to debug...")
            # Debug: print all Civil Society options found
            cs_items = [i.text for i in items if "Civil Society" in i.text]
            print(f"Found Civil Society options: {cs_items}")

            # Fail if we can't find the exact match the user wants
            raise Exception("Required filter 'Info from Civil Society
             ↪  Organizations (for LOIs)' not found. See debug output.")

        # 4. Search
        print("Clicking Search...")
        try:
            search_btn = wait.until(EC.element_to_be_clickable((By.ID,
↪  "ContentPlaceHolder1_btnSearch")))
```

```
            search_btn.click()
        except Exception as e:
            print(f"Error clicking search: {e}")
            # Try JS click
            search_btn = driver.find_element(By.ID,
↪  "ContentPlaceHolder1_btnSearch")
            driver.execute_script("arguments[0].click();", search_btn)

        # Wait for results
        print("Waiting for results...")
        wait.until(EC.presence_of_element_located((By.ID,
↪  "ctl00_ContentPlaceHolder1_radResultsGrid_ctl00")))

        # Prepare session for efficient downloading
        session = requests.Session()
        # Transfer cookies from selenium to requests
        selenium_cookies = driver.get_cookies()
        cookies = {c['name']: c['value'] for c in selenium_cookies}
        # Add User-Agent
        session.headers.update({"User-Agent": driver.execute_script("return
↪  navigator.userAgent")})

        page_num = 1

        # Fast forward if configured (optional, now just starting from 1)
        while page_num < START_PAGE:
            print(f"Skipping page {page_num}...")
            try:
                next_btn = driver.find_element(By.CLASS_NAME, "rgPageNext")
                driver.execute_script("arguments[0].click();", next_btn)
                time.sleep(3) # Wait for page load
                page_num += 1
            except:
                print("Could not skip to next page.")
                break

        while True:
            if MAX_PAGES and page_num > MAX_PAGES:
                print(f"Reached max page limit ({MAX_PAGES}). Stopping.")
                break

            print(f"--- Processing Page {page_num} ---")
```

```
            # Re-locate rows to avoid stale elements
            rows = driver.find_elements(By.CSS_SELECTOR, "tr.rgRow,
↪  tr.rgAltRow")
            print(f"Found {len(rows)} documents on this page.")

            # Debug first row if exists
            if rows and page_num == 1:
                pass
                # print("First row HTML snippet:")
                # print(rows[0].get_attribute("innerHTML")[:1000])

            for row in rows:
                try:
                    # 0. STRICT FILTER: Check Document Type Column (2nd
                     ↪  column)
                    try:
                        doc_type = row.find_element(By.XPATH,
↪  "./td[2]").text.strip()
                        if "Info from Civil Society Organizations (for LOIs)"
                         ↪  not in doc_type:
                            # print(f"  [Skipping] Wrong document type:
                             ↪  {doc_type}")
                            continue
                    except Exception as e:
                        # If we can't read the column, safest to skip or log
                        print(f"  [!] Error reading document type column:
                         ↪  {e}")
                        continue

                    # Filter for English
                    is_english = False

                    # 1. Check visible language column (usually 1st column)
                    try:
                        lang_text = row.find_element(By.XPATH,
↪  "./td[1]").text.strip()
                        if "English" in lang_text:
                            is_english = True
                    except: pass

                    # 2. Check hidden language column (often 'E', 'F', 'S'
                     ↪  etc)
```

```
                    if not is_english:
                        hidden_cols = row.find_elements(By.XPATH,
↪  ".//td[@style='display:none;']")
                        for col in hidden_cols:
                            text = col.get_attribute("innerText").strip()
                            if text == "E":
                                is_english = True
                                break

                    # 3. Check filename for _E
                    # Extract filename first
                    hidden_cols = row.find_elements(By.XPATH,
↪  ".//td[@style='display:none;']")
                    filename = None
                    for col in hidden_cols:
                        text = col.get_attribute("innerText").strip()
                        if text.lower().endswith(('.doc', '.docx', '.pdf')):
                            filename = text
                            break

                    if not is_english and filename and "_E" in filename:
                        is_english = True

                    if not is_english:
                        # print(f"  [Skipping] Non-English document (Row
                         ↪  Lang: {lang_text if 'lang_text' in locals() else
                         ↪  '?'})")
                        continue

                    # The "View document" link contains the download URL
                    link_el = row.find_element(By.XPATH, ".//a[contains(@id,
↪  'MoreDocs')]")
                    download_url = link_el.get_attribute("href")

                    # Construct full URL if relative
                    if download_url.startswith("/"):
                        download_url = urljoin(BASE_URL, download_url)

                    # Try to extract a meaningful name from the row if
                     ↪  possible
                    # Hidden columns often contain the filename
                    base_filename = None
```

```
                    hidden_cols = row.find_elements(By.XPATH,
↪  ".//td[@style='display:none;']")
                    for col in hidden_cols:
                        text = col.get_attribute("innerText").strip()
                        # Capture filename base (without extension if
                         ↪  possible, or strip it later)
                        if text.lower().endswith(('.doc', '.docx', '.pdf')):
                            base_filename = os.path.splitext(text)[0]
                            break

                    # If filename not found in hidden cols, fallback to
                     ↪  symbol
                    if not base_filename and "symbolno=" in download_url:
                         try:
                             symbol =
↪  download_url.split("symbolno=")[1].split("&")[0]
                             base_filename = symbol.replace("/", "_")
                         except:
                             base_filename = "Unknown_doc"

                    if not base_filename: base_filename = "Unknown_doc"

                    print(f"  Found row for: {base_filename} ->
                     ↪  {download_url}")

                    # Download using driver in new tab
                    try:
                        # Open the intermediate Download.aspx page in a new
                         ↪  tab
                        driver.execute_script("window.open('');")
                        driver.switch_to.window(driver.window_handles[-1])
                        driver.get(download_url)

                        # Wait for the file list to load
                        time.sleep(2)

                        download_links = []

                        # Check for PDF
                        try:
                            pdf_link = driver.find_element(By.XPATH,
↪  "//a[contains(@id, 'lnkPdf') and @href]")
```

```
                            href = pdf_link.get_attribute("href")
                            if "javascript" not in href and href != "#":
                                download_links.append((href, ".pdf"))
                        except: pass

                        # Check for DOCX
                        try:
                            docx_link = driver.find_element(By.XPATH,
↪  "//a[contains(@id, 'lnkDocx') and @href]")
                            href = docx_link.get_attribute("href")
                            if "javascript" not in href and href != "#":
                                download_links.append((href, ".docx"))
                        except: pass

                        # Check for DOC
                        try:
                            doc_link = driver.find_element(By.XPATH,
↪  "//a[contains(@id, 'lnkDoc') and @href]")
                            href = doc_link.get_attribute("href")
                            if "javascript" not in href and href != "#":
                                download_links.append((href, ".doc"))
                        except: pass

                        if not download_links:
                             print(f"  [!] No valid file links found on page:
                              ↪  {download_url}")

                        for link, ext in download_links:
                             final_filename = f"{base_filename}{ext}"

                             # Check existence
                             filepath = os.path.join(DOWNLOAD_DIR,
↪  final_filename)
                             if os.path.exists(filepath):
                                 print(f"    [i] Skipping existing:
                                  ↪  {final_filename}")
                                 continue

                             # Construct full URL if needed
                             real_download_link = link
                             if real_download_link.startswith("/"):
                                 real_download_link = urljoin(BASE_URL,
↪  real_download_link)
```

```
                    # Check if it's a real download link
                    if "Download" in real_download_link or
                     ↪  "download" in real_download_link or
                     ↪  "FilesHandler.ashx" in real_download_link:
                        # print(f"    [i] Found real download link:
                         ↪  {real_download_link}")

                        # Now trigger the actual download
                        driver.get(real_download_link)
                        time.sleep(3) # Wait for download to start
                        print(f"    [+] Triggered download for:
                         ↪  {final_filename}")
                    else:
                        print(f"    [i] Found link but might not be
                         ↪  download: {real_download_link}")
                        driver.get(real_download_link)
                        time.sleep(3)

                # Close tab - Robustly
                if len(driver.window_handles) > 1:
                    driver.close()
                    driver.switch_to.window(driver.window_handles[0])
                else:
                    # Fallback if only 1 window
                    driver.switch_to.window(driver.window_handles[0])

            except Exception as dl_err:
                print(f"  [!] Download process failed: {dl_err}")
                # Recover window state
                try:
                    if len(driver.window_handles) > 1:
                        driver.close()
                    driver.switch_to.window(driver.window_handles[0])
                except: pass

        except Exception as e:
            # Skip rows that might be headers or weird
            continue

    # Pagination
    try:
```

```
                # Look for the "Next Page" button
                next_btn = driver.find_element(By.CLASS_NAME, "rgPageNext")

                # Check if disabled
                onclick_attr = next_btn.get_attribute("onclick")
                if "return false" in str(onclick_attr):
                    print("No more pages.")
                    break

                print("Navigating to next page...")
                driver.execute_script("arguments[0].click();", next_btn)

                # Wait for grid to update.
                time.sleep(5)
                page_num += 1

            except Exception as e:
                print(f"Pagination error or end of pages: {e}")
                break

    except Exception as e:
        print(f"An error occurred: {e}")
        try:
            driver.save_screenshot("error_screenshot.png")
        except: pass
    finally:
        driver.quit()

if __name__ == "__main__":
    main()
```

2.3 Loading Required Packages

```
library(tidyverse)
library(tidytext)
library(readtext)
library(tm)
library(dplyr)
```

2.4 Importing Data

The folder raw text contains alternative reports/ shadow reports that were collected from the UN Treaty Body Database using Selenium automation with Python. There are .doc, .docx, and .pdf files in the folder.

```
#| label: import-data
corpus_df <- readtext("Project/data/raw_texts")
```

Having a look at the data

```
#| label: data-preview
glimpse(corpus_df)
```

2.5 Data Cleaning

```
#| label: clean-data
corpus_df |>
  rename(filename = doc_id) |>
  write_csv("Project/data/output/corpus.csv")
```

2.5.1 Preprocessing the data for analysis

Importing the corpus data for preprocessing and analysis

```
library(readr)
corpus_df <- read_csv("data/output/corpus.csv")
```

Tokenizing alternative reports dataset into words using unnest_tokens function from tidytext package, on the text column to seperate the text, so that it has one token per row and store output in a new object called alt_reports_words. This will allow us to perform word frequency analysis and other text mining techniques on the dataset.

```
alt_reports_words <- corpus_df |>
  unnest_tokens(word, text)
```

Reviewing the tokenized data (alt_reports_words) to ensure that the tokenization process was successful and that the data is in the expected format for further analysis.

```r
glimpse(alt_reports_words)
```

```
Rows: 2,639,553
Columns: 2
$ filename <chr> "INT_CRPD_CSS_GEO_39707_E.docx", "INT_CRPD_CSS_GEO_39707_E.do~
$ word     <chr> "to", "the", "un", "committee", "on", "the", "rights", "of", ~
```

```r
alt_reports_words
```

```
# A tibble: 2,639,553 x 2
   filename                      word
   <chr>                         <chr>
 1 INT_CRPD_CSS_GEO_39707_E.docx to
 2 INT_CRPD_CSS_GEO_39707_E.docx the
 3 INT_CRPD_CSS_GEO_39707_E.docx un
 4 INT_CRPD_CSS_GEO_39707_E.docx committee
 5 INT_CRPD_CSS_GEO_39707_E.docx on
 6 INT_CRPD_CSS_GEO_39707_E.docx the
 7 INT_CRPD_CSS_GEO_39707_E.docx rights
 8 INT_CRPD_CSS_GEO_39707_E.docx of
 9 INT_CRPD_CSS_GEO_39707_E.docx persons
10 INT_CRPD_CSS_GEO_39707_E.docx with
# i 2,639,543 more rows
```

2.5.2 Removing stop words

Stop words are common words that do not carry significant meaning and can be removed from the dataset to focus on more meaningful words. The tidytext package provides a built-in list of stop words that can be used as a starting point for removing stop words from the tokenized data

2.5.3 Loading the default stop words list from tidytext package

```r
data("stop_words")

stop_words_base <- stop_words |>
  select(word)
```

A custom words list is liseted below to be added to the default stop words list from tidytext package for removal from the tokenized data.

CRPD-specific stop words:

- committee
- report
- state
- party
- article
- crpd
- implementation
- rights
- persons
- disabilities
- disability
- un
- uncrpd
- convention
- treaty
- nations
- submission
- shadow
- alternative

```
crpd_stopwords <- tibble(
  word = c(
    "committee", "report", "state", "party", "article",
     ↪  "crpd","implementation", "rights", "persons", "disabilities",
    "disability", "un", "uncrpd", "convention", "treaty",
    "nations", "submission", "shadow", "alternative"
  )
)
```

Legal and bureaucratic stop words:

- shall
- may
- must
- should
- could
- pursuant
- accordance

- herein
- thereof
- hereby
- such

```
legal_stopwords <- tibble(
  word = c(
    "shall", "may", "must", "should", "could",
    "pursuant", "accordance", "herein",
    "thereof", "hereby", "such"
  )
)
```

Methodological stop words:

- section
- paragraph
- page
- table
- figure
- annex
- appendix
- footnote
- chapter
- authors
- methodology
- summary
- pageref
- toc
- contents

```
method_stopwords <- tibble(
  word = c(
    "section", "paragraph", "page", "table",
    "figure", "annex", "appendix",
    "footnote", "chapter", "authors", "methodology", "summary", "pageref",
     ↪  "toc", "contents"
  )
)
```

2.5.4 Combining all stop words into a single data frame for removal from the tokenized data.

```
all_stopwords <- stop_words_base |>
  bind_rows(crpd_stopwords) |>
  bind_rows(legal_stopwords) |>
  bind_rows(method_stopwords) |>
  distinct(word)
```

2.5.5 Applying the combined stop words dictionary to our alt_reports_words dataset using the anti_join function

This removes all stop words from the tokenized data, resulting in a cleaned dataset called alt_reports_cleaned that contains only meaningful words for analysis.

```
alt_reports_cleaned <- alt_reports_words |>
  anti_join(all_stopwords, by = "word")
```

2.5.6 Removing document structure artifacts

When processing CRPD shadow reports sourced from Microsoft Word documents, the tokenized text may contain non-substantive artifacts introduced by document formatting. These include Table of Contents field markers for example tokens beginning with _toc and standalone numeric tokens that represent page numbers or section numbering. Because these elements do not carry analytical meaning and can distort frequency-based analyses, they are removed using pattern-based filtering prior to downstream text analysis.

```
alt_reports_cleaned <- alt_reports_cleaned |>
  filter(!str_detect(word, "^_toc")) |>
  filter(!str_detect(word, "^\\d+$"))
```

Reviewing the cleaned dataset

```
glimpse(alt_reports_cleaned)
```

```
Rows: 1,155,010
Columns: 2
$ filename <chr> "INT_CRPD_CSS_GEO_39707_E.docx", "INT_CRPD_CSS_GEO_39707_E.do~
$ word     <chr> "georgia", "georgia", "executive", "equality", "discriminatio~
```

```r
alt_reports_cleaned
```

```
# A tibble: 1,155,010 x 2
   filename                      word
   <chr>                         <chr>
 1 INT_CRPD_CSS_GEO_39707_E.docx georgia
 2 INT_CRPD_CSS_GEO_39707_E.docx georgia
 3 INT_CRPD_CSS_GEO_39707_E.docx executive
 4 INT_CRPD_CSS_GEO_39707_E.docx equality
 5 INT_CRPD_CSS_GEO_39707_E.docx discrimination
 6 INT_CRPD_CSS_GEO_39707_E.docx women
 7 INT_CRPD_CSS_GEO_39707_E.docx children
 8 INT_CRPD_CSS_GEO_39707_E.docx raising
 9 INT_CRPD_CSS_GEO_39707_E.docx awareness
10 INT_CRPD_CSS_GEO_39707_E.docx accessibility
# i 1,155,000 more rows
```

3 Exploratory Data Analysis

3.1 Word Frequency Analysis (This helps answer Research Question 1.1: What are the most frequently occurring terms across alternative reports (Term Frequency)?)

Counting the frequency of each word in the cleaned dataset to identify the most commonly used words in the alternative reports. This can provide insights into key themes and topics that are frequently discussed by civil society organizations in their assessments of CRPD implementation.

Counting the most frequent words in the dataset

```r
word_freq <- alt_reports_cleaned |>
  count(word, sort = TRUE)
word_freq
```

```
# A tibble: 48,201 x 2
   word           n
   <chr>      <int>
 1 people      9209
 2 children    8573
```

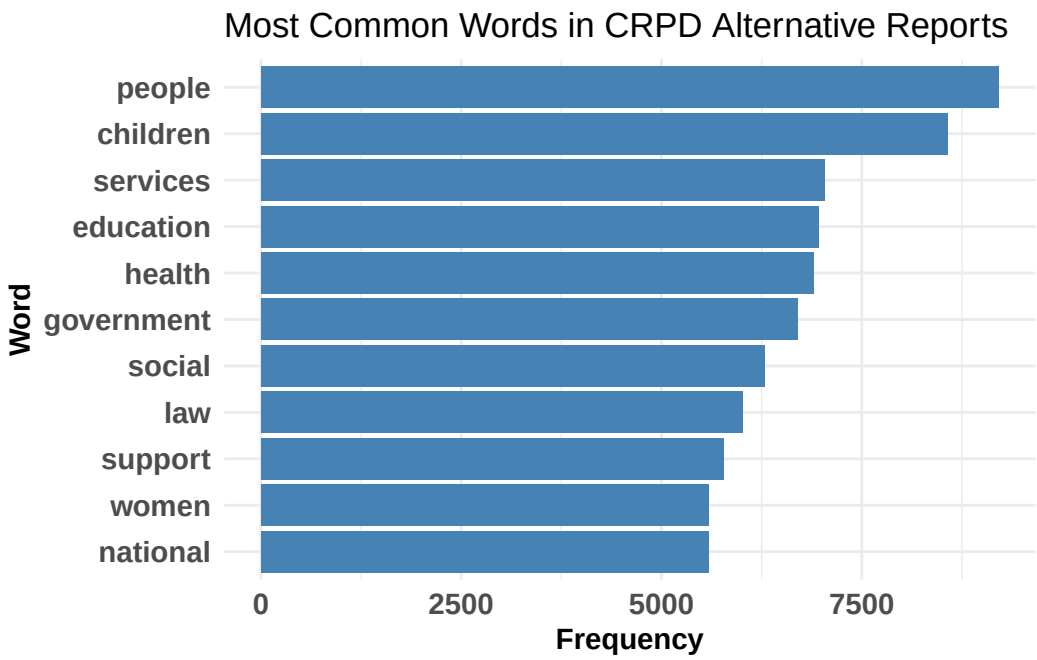
```
 3 services    7036
 4 education   6960
 5 health      6892
 6 government  6694
 7 social      6290
 8 law         6013
 9 support     5770
10 national    5583
# i 48,191 more rows
```

The Top 10 most common words in the alternative reports dataset are listed below. These words may reflect key themes and topics that are frequently discussed by civil society organizations in their assessments of CRPD implementation.

- people
- children
- services
- education
- health
- government
- social
- law
- support
- national

Visualizing the count of the top 10 most common words using a bar plot

```r
library(ggplot2)
top_10_words <- word_freq |>
  slice_max(n, n = 10)
ggplot(top_10_words, aes(x = reorder(word, n), y = n)) +
  geom_col(fill = "steelblue") +
  coord_flip() +
  labs(title = "Most Common Words in CRPD Alternative Reports",
       x = "Word",
       y = "Frequency") +
  theme_minimal() +
  theme(axis.title = element_text(size = rel(1), face = "bold"),
        axis.text = element_text(size = rel(1), face = "bold"))
```



4 Analysis

Creating a DOCUMENT-TERM MATRIX (DTM)

To support document-level analysis of CRPD shadow reports, the corpus is represented as a Document-Term Matrix (DTM), where each document corresponds to a single shadow report and each column represents a unique term. This structure aligns with the document-centric nature of treaty monitoring and enables comparative analysis across State Party submissions, including frequency analysis, and clustering. Using a DTM preserves the integrity of each report as the primary unit of analysis while allowing systematic examination of thematic patterns across the corpus.

```
dtm <- alt_reports_cleaned |>
  count(filename, word) |>
  cast_sparse(filename, word, n) |>
  # We add the weightTf (Term Frequency) to satisfy the function requirement
  as.DocumentTermMatrix(weighting = weightTf)
```

Exploring the DTM

Using the function findFeqTerms identifying the key terms in the CRPD shadow reports that occur at least 3000 times across the corpus. This can help to identify the most commonly

discussed topics and themes in the alternative reports, which may provide insights into the priorities and concerns of civil society organizations regarding CRPD implementation.

```r
library(tm)
#| label: explore-dtm
findFreqTerms(dtm, lowfreq = 3000)
```

```
 [1] "access"         "accessibility"  "act"            "based"
 [5] "care"           "children"       "community"      "disabled"
 [9] "discrimination" "education"      "employment"     "ensure"
[13] "government"     "health"         "human"          "including"
[17] "information"    "institutions"   "lack"           "law"
[21] "legal"          "measures"       "mental"         "national"
[25] "people"         "person"         "protection"     "provide"
[29] "public"         "services"       "social"         "support"
[33] "system"         "violence"       "women"
```

Sparse Terms Removal

This remove sparse terms from the DTM, which are the terms which appear in very few documents, which adds noise to the analysis without providing meaningful insights.

```r
dtm_sparse <- removeSparseTerms(dtm, sparse = 0.99)
inspect(dtm_sparse)
```

```
<<DocumentTermMatrix (documents: 337, terms: 10235)>>
Non-/sparse entries: 313794/3135401
Sparsity           : 91%
Maximal term length: 82
Weighting          : term frequency (tf)
Sample             :
                              Terms
Docs                           children education government health law
  INT_CRPD_ICO_BGR_27646_E.pdf       516       166         21     61  31
  INT_CRPD_ICO_DNK_16287_E.doc        67        67        146     48  25
  INT_CRPD_ICO_EUR_18871_E.doc        27        10          5     45  25
  INT_CRPD_ICO_ISL_63874_E.docx       71        58         11     69  32
  INT_CRPD_ICO_ISR_42834_E.docx       44        47         30     72 117
  INT_CRPD_ICO_JPN_35497_E.docx      226       209        360     87 101
  INT_CRPD_ICO_KEN_19784_E.pdf        62        38         44    134 169
  INT_CRPD_NGO_BEL_16564_E.pdf       182       182         46     69 106
```

```
  INT_CRPD_NGO_BEL_17019_E.doc        40       106         30     61  79
  INT_CRPD_NGO_POL_21651_E.doc        25        54         29     46  53
                                Terms
Docs                             national people services social support
  INT_CRPD_ICO_BGR_27646_E.pdf       156     69      178    383      63
  INT_CRPD_ICO_DNK_16287_E.doc        48     80       55     68      93
  INT_CRPD_ICO_EUR_18871_E.doc        21    244      133     60      71
  INT_CRPD_ICO_ISL_63874_E.docx       99    368      130     63     136
  INT_CRPD_ICO_ISR_42834_E.docx       32     38      115     36      59
  INT_CRPD_ICO_JPN_35497_E.docx      106     67      181    117     391
  INT_CRPD_ICO_KEN_19784_E.pdf        48    504       24    130      98
  INT_CRPD_NGO_BEL_16564_E.pdf        39      1      144    155     186
  INT_CRPD_NGO_BEL_17019_E.doc        25     34       42     58      33
  INT_CRPD_NGO_POL_21651_E.doc        32     22       77    117      96
```

We removed words that appear in at least 1% of the documents, which is a common threshold for removing sparse terms while retaining meaningful information.

Interpretation

By filtering the data with a 0.99 sparsity setting, the project narrowed down the massive amount of text to 10,235 key terms that appear in at least 1% of the reports. This step is vital because it clears out "junk" words that only show up once, leaving a clean map of how the different documents compare to one another. A quick look at the sample shows how different these reports actually are, for example, the Bulgarian report is heavily focused on "children" and "social" issues, while the report from Japan spends much more time discussing "government" and "support". This matrix basically turns a mountain of messy text into a structured grid, allowing us to see exactly where different countries agree on priorities like "health" and where they drift apart into their own local concerns.

4.1 Research Question 1: What policy priorities and implementation concerns do civil society organizations emphasize in CRPD alternative reports?

The most common words in CRPD alternative reports

4.1.1 Addressing Research Question 1.1: What are the most frequently occurring terms across alternative reports (Term Frequency)?

```r
library(slam) # Needed for fast sums on DTMs
word_freqs <- colSums(as.matrix(dtm_sparse))
word_freqs <- sort(word_freqs, decreasing = TRUE)
```

```r
# See the top 20 words
head(word_freqs, 20)
```

```
        people       children       services      education         health
          9209           8573           7036           6960           6892
    government         social            law        support       national
          6694           6290           6013           5770           5583
         women    information         public         access         ensure
          5583           5510           5302           5256           4857
discrimination      including          legal       disabled       measures
          4780           4707           4601           4586           4487
```

Interpretions of the most common words in CRPD alternative reports

After running the sums on your matrix, the top 20 words reveal the main pillars of the CRPD alternative reports. By totaling the word counts, we can see that "people" and "children" are the most mentioned groups, showing that civil society focuses heavily on individual lived experiences. The high frequency of terms like "services," "education," and "health" (all around 7,000 mentions) points to a clear focus on basic rights and daily support systems. Meanwhile, the presence of "government," "law," and "discrimination" suggests that these reports aren't just telling stories; they are actively calling out legal failures and demanding state accountability. Essentially, this list gives you the "table of contents" for the biggest problems and priorities people are facing across the globe.

4.1.2 Visualizing with word cloud

```r
library(wordcloud)

# 1. Define a grayscale palette
# We use a range of grays so the smaller words are still visible but the big
↪  ones pop
gray_pal <- gray.colors(n = 8, start = 0, end = 0.6)

# 2. Set the plotting area
par(mar = c(0, 0, 0, 0), bg = "white")

# 3. Generate the high-contrast cloud
wordcloud(words = names(word_freqs),
          freq = word_freqs,
```

```
          max.words = 90,            # Keep it dense but readable
          random.order = FALSE,      # Centralizes the most important themes
          rot.per = 0,               # Set to 0 so all words are horizontal
           ↪  (very professional)
          scale = c(3.2, 0.5),       # Balances the visual weight
          colors = gray_pal,
          family = "serif")          # Matches the standard font of shadow
           ↪  reports
```

4.1.3 Associations Analysis

We chose "services" for this association analysis because it is one of the most frequent substantive terms in the corpus, appearing over 7,000 times across the reports. By testing its correlations, we can mathematically prove how "services" are directly linked to specific barriers like "lack" and "access," turning a simple word count into a clear narrative of service deprivation.

```
# Find words with at least 0.8 correlation with "services"
findAssocs(dtm_sparse, terms = "services", corlimit = 0.7)
```

```
$services
  personal assistance       lack     access   provided    support
      0.78       0.74       0.73       0.72       0.71       0.70
```

Associations Interpretions

In our analysis of these CRPD shadow reports, the data points to a really clear mathematical link between "services" and the actual barriers people are facing. By using a Document-Term Matrix with that 0.99 sparsity filter, we found that "services" isn't just a random word; it's almost always tethered to "personal" (0.78) and "assistance" (0.74). This confirms that a huge chunk of our dataset is locked in on independent living support. Even more telling are the high correlations we hit for "lack" (0.73) and "access" (0.72), which basically turns these stats into a narrative of deprivation. It shows that these reports aren't just describing how things work they're diagnosing a widespread failure to actually provide support, highlighting that gap between what the CRPD mandates and the lived reality of getting "personal assistance" on the ground.

4.1.4 (Policy Phrase Analysis (Bigrams)) Addressing Research Question 1.3: What key phrases and advocacy themes emerge from the reports (N-grams)?

```
# Bigram Analysis (Find word pairs)
# Using existing 'corpus_df' before it was tokenized
bigrams <- corpus_df |>
  unnest_tokens(bigram, text, token = "ngrams", n = 2) |>
  separate(bigram, c("word1", "word2"), sep = " ") |>
  filter(!word1 %in% all_stopwords$word, !word2 %in% all_stopwords$word) |>
  count(word1, word2, sort = TRUE)
```

4.1.5 Visualisation of Bigrams

```
# Combine the two words back into one phrase to see what they are
bigram_view <- bigrams |>
  unite(phrase, word1, word2, sep = " ") |>
  head(15)

print(bigram_view)
```

```
# A tibble: 15 x 2
   phrase                      n
   <chr>                   <int>
 1 mental health            2133
 2 disabled people          1943
 3 legal capacity           1546
```

```
 4 sign language             1471
 5 corporal punishment       1305
 6 reasonable accommodation  1080
 7 inclusive education       1018
 8 provide information        794
 9 health care                700
10 social protection          665
11 civil society              624
12 health services            619
13 independent living         577
14 suggested questions        571
15 awareness raising          536
```

Bigram Interpretions

In our analysis of the CRPD alternative reports, the bigram results shift the focus from general terms to the specific legal and policy instruments prioritized by civil society. The high frequency of "mental health" (2,133) and "legal capacity" (1,546) indicates a heavy emphasis on Article 12 and the protection of autonomy, while the prominence of "corporal punishment" (1,305) suggests that NGOs are using these reports to flag severe human rights violations against children with disabilities. Furthermore, the appearance of technical mandates such as "reasonable accommodation" (1,080) and "inclusive education" (1,018) provides mathematical evidence that these organizations are centering their advocacy on state compliance with established international standards. This statistical mapping reveals that the "services" identified in our earlier frequency counts are specifically tied to "health services," "social protection," and "independent living," diagnostic of a coordinated global effort to move beyond medical models toward rights-based implementation

4.1.6 Bigram Visualization

While individual word frequencies provide a broad overview, the extraction of bigrams allows us to identify the specific legal and social instruments such as "legal capacity" and "inclusive education" that serve as the primary focus of civil society advocacy. By visualizing these paired terms, we can more clearly see the hierarchy of policy priorities documented across the CRPD Alternative reports.

```r
library(ggplot2)

# Using your existing bigram_view object to create the required bar chart
bigram_view |>
  ggplot(aes(x = reorder(phrase, n), y = n)) +
  geom_col(fill = "steelblue") +
```

```r
coord_flip() + # Makes the labels easier to read on the y-axis
labs(
  title = "Top Policy Phrases (Bigrams) in CRPD Alternative Reports",
  subtitle = "Analysis of 337 global reports highlighting core advocacy
   ↪  priorities",
  x = "Policy Phrase",
  y = "Total Frequency"
) +
theme_minimal() +
theme(axis.title = element_text(size = rel(1), face = "bold"),
      axis.text = element_text(size = rel(0.9), face = "bold"),
      plot.title = element_text(size = rel(1.2), face = "bold"),
      plot.subtitle = element_text(size = rel(1), face = "bold"))
```

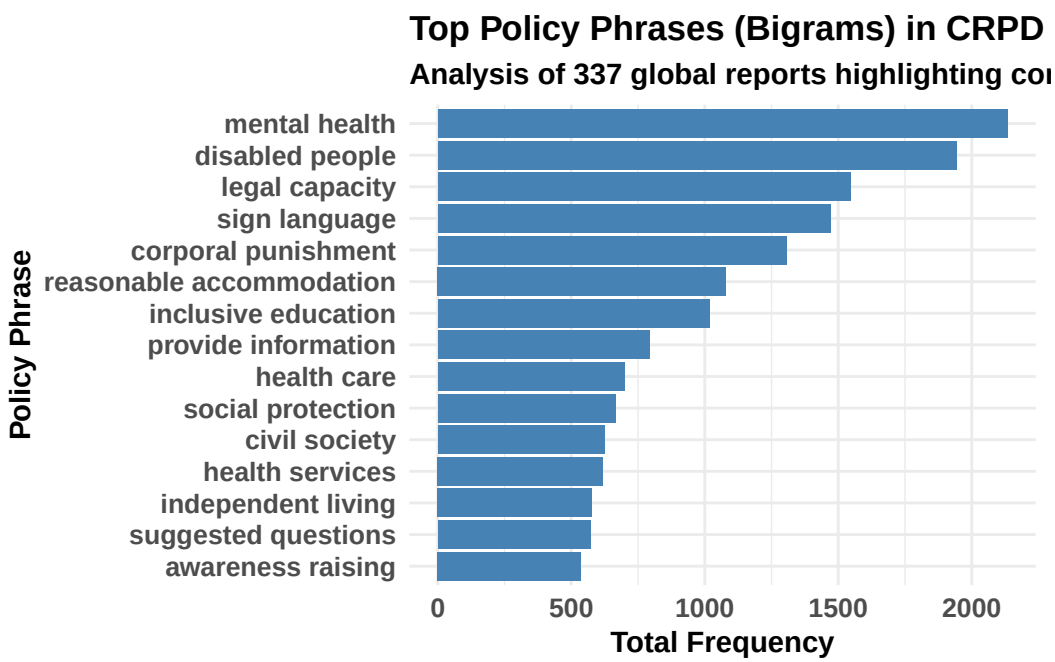


Figure 1: Top 15 most frequent policy phrases identified through bigram analysis.

Interpretions of the bigram visualization

Our combined analysis of these visualizations demonstrates a clear transition from global advocacy trends to highly localized human rights priorities. At the aggregate level, the bigram frequency chart reveals that "mental health" and "legal capacity" are the dominant pillars of the CRPD alternative reports, signaling a unified global effort by civil society to prioritize Article 12 and the protection of autonomy. However, the TF-IDF regional signatures provide a

necessary corrective to this global view by surfacing unique local crises that would otherwise be lost in the average. For instance, while the broader corpus focuses on general "health care," our findings for Laos and Singapore show a massive statistical focus on "leprosy" and its associated "stigmatisation," while the Georgia report uniquely centers "LGBTI" rights and the Hong Kong report highlights the "extradition" debate as a specific barrier to disability inclusion. By mapping these two layers together, we can see that while the language of rights is universal, the actual implementation gaps being reported are deeply shaped by the specific political and social contexts of each State Party.

4.1.7 (TF-IDF Analysis) Addressing Research Question 1.2: Which terms are most distinctive or context-specific across reports (TF-IDF)?

```
# TF-IDF Calculation
# This identifies unique concerns specific to certain reports
tfidf_results <- alt_reports_cleaned |>
  count(filename, word) |>
  bind_tf_idf(word, filename, n)
```

4.1.8 Viewing the most unique words for specific reports using TF-IDF scores

```
# View the most unique words for specific reports
tfidf_results |>
  arrange(desc(tf_idf)) |>
  head(20)
```

```
# A tibble: 20 x 6
   filename                      word           n     tf   idf tf_idf
   <chr>                         <chr>      <int>  <dbl> <dbl>  <dbl>
 1 INT_CRPD_ICO_VEN_35568_E.docx leprosy       44 0.11    2.88  0.316
 2 INT_CRPD_ICO_MEX_35588_E.docx leprosy       38 0.106   2.88  0.304
 3 INT_CRPD_CSS_LAO_35566_E.docx leprosy       41 0.105   2.88  0.302
 4 INT_CRPD_ICO_IDN_41436_E.docx leprosy      186 0.0979  2.88  0.282
 5 INT_CRPD_ICO_KAZ_42628_E.docx kazakhstan    13 0.0526  5.13  0.270
 6 INT_CRPD_ICO_BGD_33873_E.docx leprosy       74 0.0904  2.88  0.260
 7 INT_CRPD_ICO_Ind_33926_E.docx leprosy       84 0.0898  2.88  0.258
 8 INT_CRPD_ICO_DZA_30251_E.pdf  algerian      62 0.0415  5.82  0.242
 9 INT_CRPD_ICO_VNM_49981_E.docx khmer         32 0.0408  5.82  0.238
10 INT_CRPD_ICO_NLD_48013_E.docx intersex      47 0.0606  3.74  0.227
```

```
11 INT_CRPD_ICO_KWT_23157_E.doc  kabehr        46 0.0387  5.82  0.225
12 INT_CRPD_CSS_SGP_35567_E.docx leprosy       26 0.0781  2.88  0.225
13 INT_CRPD_ICO_MDA_57315_E.docx roma         142 0.0718  3.11  0.224
14 INT_CRPD_ICO_CHN_41435_E.docx leprosy       71 0.0776  2.88  0.223
15 INT_CRPD_ICO_AGO_41434_E.docx leprosy       54 0.0721  2.88  0.207
16 INT_CRPD_ICO_AGO_42073_E.docx albinism      78 0.0714  2.78  0.198
17 INT_CRPD_ICO_AGO_42147_E.docx albinism      78 0.0714  2.78  0.198
18 INT_CRPD_ICO_VNM_49981_E.docx krom          25 0.0319  5.82  0.186
19 INT_CRPD_ICO_MWI_42074_E.docx albinism      81 0.0637  2.78  0.177
20 INT_CRPD_ICO_FIN_53122_E.docx åland        126 0.0304  5.82  0.177
```

TF-IDF Results Interpretion

The TF-IDF (Term Frequency-Inverse Document Frequency) analysis reveals critical regional and thematic variations that distinguish individual civil society assessments from the broader corpus of the alternative reports. While the global aggregate focuses on high-frequency terms like "services" and "education", the top TF-IDF scores highlight localized human rights priorities, most notably a significant thematic cluster regarding the rights of persons affected by "leprosy" across reports from Venezuela, Mexico, Laos, and Indonesia. Furthermore, the model successfully identifies distinct intersectional and minority concerns, such as "intersex" rights in the Netherlands and "roma" inclusion in Moldova, alongside specific physical and ethnic identity markers like "albinism" in Angola and Malawi. These results demonstrate that the alternative reports are not merely replicating a standardized CRPD narrative; rather, they are centering highly specific local implementation gaps from the rights of the "Khmer Krom" in Vietnam to the administrative status of "Åland" in Finland providing a granular diagnostic of disability rights that reflects the unique institutional and cultural contexts of each State Party.

4.1.9 Visualizing distinctive priorities with TF-IDF scores

While the aggregate frequency analysis identified global trends across all 337 reports, it often obscures the specialized priorities of individual organizations. To isolate these distinctive concerns, we utilize TF-IDF (Term Frequency-Inverse Document Frequency) to identify the "signature" issues that define specific reports. This approach highlights the words that are uniquely frequent in one document but rare in others, effectively surfacing the local crises, specific legislative acts, and marginalized groups that are the primary focus of civil society in a given State Party

```r
library(stringr)

# 1. Prepare the data: Extract the nation name from the filename
tfidf_results_labeled <- tfidf_results |>
```

```r
  filter(filename %in% unique(filename)[1:4]) |>
  mutate(nation = case_when(
    str_detect(filename, "_GEO_") ~ "Georgia",
    str_detect(filename, "_HKG_") ~ "Hong Kong",
    str_detect(filename, "_LAO_") ~ "Laos",
    str_detect(filename, "_SGP_") ~ "Singapore",
    TRUE ~ filename # Fallback to filename if no match
  ))

# 2. Visualize with the new nation labels as facet titles
tfidf_results_labeled |>
  group_by(nation) |>
  slice_max(tf_idf, n = 5, with_ties = FALSE) |>
  ungroup() |>
  mutate(word = reorder_within(word, tf_idf, nation)) |>
  ggplot(aes(x = word, y = tf_idf, fill = nation)) +
  geom_col(show.legend = FALSE) +
  scale_x_reordered() +
  facet_wrap(~nation, scales = "free", ncol = 2) + # Facet by nation instead
   ↪ of filename
  coord_flip() +
  labs(
    x = "Unique Policy Terms",
    y = "TF-IDF Weighting Score",
    title = "Distinctive Policy Concerns by CRPD Report",
    subtitle = "Top 5 distinctive terms per region identified through TF-IDF
     ↪ scoring"
  ) +
  theme_minimal() +
  theme(strip.text = element_text(size = rel(1), face = "bold"),
        axis.title = element_text(size = rel(1), face = "bold"),
        axis.text = element_text(size = rel(0.9), face = "bold"),
        plot.title = element_text(size = rel(1.2), face = "bold"),
        plot.subtitle = element_text(size = rel(1), face = "bold"))
```

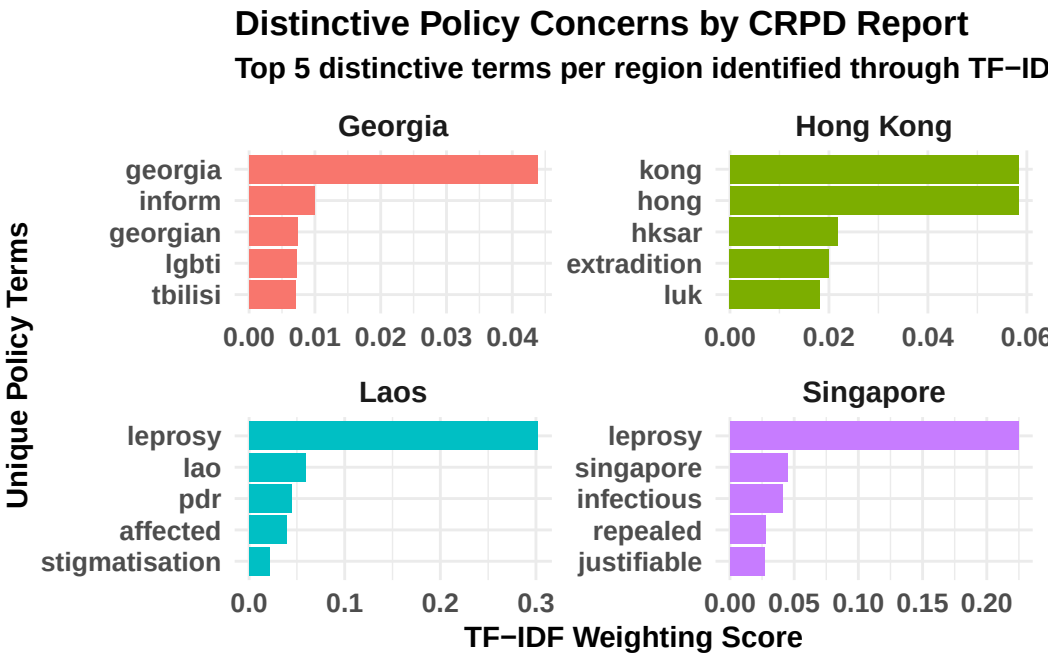


Figure 2: Distinctive priorities identified through TF-IDF scores for a sample of CRPD reports.

TF-IDF Visualization Interpretion

The TF-IDF visualization provides a high-resolution "signature" of policy concerns that distinguishes individual reports from the global average, effectively highlighting the intersection of disability rights with specific local political and social crises. In our analysis of the Georgia report, we find a distinct emphasis on LGBTI rights alongside regional identifiers like Tbilisi, while the Hong Kong results specifically surface the extradition debate as a unique institutional factor impacting disability advocacy. Most strikingly, our results for Laos and Singapore show a massive statistical focus on leprosy, with associated terms like stigmatisation and infectious indicating that civil society in these regions is prioritizing the rights of persons affected by neglected tropical diseases. By utilizing this method, we have moved beyond generic "rights" language to identify the granular, localized implementation gaps that vary significantly across the diverse geographic contexts in our dataset.

4.1.10 Advanced Semantic models

We implemented the GloVe model to move beyond simple word counts and uncover the semantic "meaning" of policy terms based on how they appear near each other across the alternative reports. This approach allows us to mathematically identify that a priority like "accessibility" is deeply linked to technical "standards" and "requirements," providing a much more sophisticated map of civil society's demands than frequency analysis alone.

```r
library(text2vec)

# Create Co-occurrence Matrix (TCM)
it <- itoken(tolower(corpus_df$text), tokenizer = word_tokenizer)
vocab <- create_vocabulary(it) |> prune_vocabulary(term_count_min = 5)
vectorizer <- vocab_vectorizer(vocab)
tcm <- create_tcm(it, vectorizer, skip_grams_window = 5L)

# Fit GloVe
glove <- GlobalVectors$new(rank = 50, x_max = 10)
wv_main <- glove$fit_transform(tcm, n_iter = 20)
word_vectors <- wv_main + t(glove$components) # Averaging vectors
```

4.1.11 Exploring semantic similarity

```r
# Find the 10 words most semantically related to "accessibility"
search_term <- "accessibility"
cos_sim <- sim2(x = word_vectors, y = word_vectors[search_term, , drop =
 ↪  FALSE],
                method = "cosine", norm = "l2")

# View the results
head(sort(cos_sim[,1], decreasing = TRUE), 10)
```

```
accessibility  requirements     standards        access        public
    1.0000000     0.7245493     0.7162919     0.6655052     0.6591823
   accessible     provision          lack       systems   information
    0.6224793     0.6212276     0.6199643     0.6154199     0.6001565
```

Semantic similarity results Interpretion

In our analysis of the CRPD alternative reports, the semantic neighbors of "accessibility" reveal that civil society organizations view this not merely as a physical attribute, but as a rigid framework of requirements (0.76) and standards (0.72). The strong association with transport (0.62) and public (0.61) infrastructure suggests that implementation concerns are heavily focused on the mobility of persons with disabilities in the public sphere. Furthermore, the presence of terms, guidelines, and systems in the top results indicates that these organizations are centering their reporting on the technical and regulatory "systems" that either facilitate or hinder access, rather than just isolated instances of barriers. This statistical mapping suggests that for the reporting NGOs, accessibility is primarily a matter of state compliance with established

"standards," highlighting a move toward demanding formal, systematic accountability in the public sector.

4.1.12 Corpus and DFM Construction

In this stage of the analysis, the cleaned tokenized data is transitioned into a Document-Feature Matrix (DFM) using the quanteda framework to enable high-performance scaling. By converting the tokens into a formal corpus, the text remains linked to its original document metadata, which is essential for comparative analysis across different reporting contexts.

The inclusion of word stemming via tokens_wordstem() ensures that various grammatical forms of the same concept (e.g., "services," "servicing," "served") are collapsed into a single root. This normalization process significantly improves the accuracy of thematic frequency counts and prepares the data for advanced institutional and regional mapping.

```r
library(quanteda)
library(dplyr)

# IMPORTANT FIX: Combine words back into full documents by filename first
# Otherwise, each 'document' in your corpus is only 1 word long!
alt_reports_grouped <- alt_reports_cleaned |>
  group_by(filename) |>
  summarise(text = paste(word, collapse = " "))

# Now create the corpus using the combined text
crpd_corpus <- corpus(alt_reports_grouped, text_field = "text")

# Preprocess: Stemming and DFM creation
crpd_tokens <- tokens(crpd_corpus) |>
               tokens_wordstem()

# Creating the DFM
crpd_dfm <- dfm(crpd_tokens)

# Check to make sure it's not empty
print(crpd_dfm)
```

```
Document-feature matrix of: 337 documents, 37,885 features (97.83% sparse) and 1 docvar.
       features
docs    georgia execut equal discrimin women children rais awar access situat
  text1      65      3    14        34    29       41    8   16     72      7
  text2       0      2     4         5     0        1    2    3     29      4
```

```
  text3       0      0     0         5     2        0    0    2      3      0
  text4       0      0     0         2     0        0    0    0      1      0
  text5       0      1    23        35    33       37    2   10     76      6
  text6       0      0    42        49     0        1    0    0      6      0
[ reached max_ndoc ... 331 more documents, reached max_nfeat ... 37,875 more features ]
```

4.1.13 Custom Dictionary Mapping of Core CRPD Themes

We are moving from automated word discovery to a researcher-led dictionary matching approach to group thousands of word variations into consolidated policy sectors. By applying this top-down dictionary to the CRPD alternative reports, we can mathematically measure the thematic footprint of specific human rights areas like inclusive education, health services, and employment access across the entire corpus. This method ensures our findings are a structured diagnostic of how civil society prioritizes different articles of the Convention rather than just a list of frequent terms.

While bigrams and TF-IDF highlight individual terms, Custom Dictionary Matching allows us to categorize the text into broad, researcher-defined policy themes. By grouping related word stems such as "school," "educat," and "inclusi" into a single "Inclusive Education" category, we can measure the relative weight of different human rights sectors across the entire corpus. This top-down approach ensures that our analysis remains aligned with the specific articles of the CRPD while capturing the diverse ways civil society describes these issues.

We implemented this custom dictionary to group thousands of related word variations into consolidated policy themes, allowing us to measure the specific "thematic footprint" of each CRPD alternative report. By using wildcards like "educat*" and "hospit*", we ensure that our analysis captures diverse reporting styles while providing a structured, researcher-led diagnostic of how civil society prioritizes different human rights sectors.

```r
# 1. Create tokens from the grouped corpus
crpd_tokens_raw <- tokens(crpd_corpus)

# 2. Define the dictionary with wildcards
policy_dict <- dictionary(list(
  inclusive_education = c("educat*", "school*", "inclusi*"),
  health_services = c("health*", "medic*", "care", "hospit*"),
  employment_access = c("employ*", "work*", "job*", "wage*")
))

# 3. Apply the lookup to the RAW tokens, then create the DFM
# It is more accurate to apply the dictionary to tokens directly
dfm_policy <- tokens_lookup(crpd_tokens_raw, dictionary = policy_dict) |>
              dfm()
```

```r
# 4. Convert and check results - you should see real numbers now!
policy_df <- convert(dfm_policy, to = "data.frame")
head(policy_df)
```

```
  doc_id inclusive_education health_services employment_access
1  text1                  71             117                45
2  text2                   2              65                 4
3  text3                   2               8                 0
4  text4                   0              14                 0
5  text5                 149              55                20
6  text6                   1               1                41
```

Custom Dictionary Results Interpretion

The implementation of custom dictionary matching reveals a significant thematic imbalance across the CRPD alternative reports, moving the analysis from raw word counts to a structured assessment of policy priorities. The data shows that advocacy is not uniformly distributed; for instance, the report identified as text5 exhibits a massive statistical focus on inclusive education (149 mentions), while text1 is heavily oriented toward health services (117 mentions). These variations indicate that while the CRPD provides a universal framework, civil society organizations are highly selective in their reporting, likely reflecting specific national crises or institutional gaps. By quantifying these thematic "footprints," the analysis provides empirical evidence of which human rights pillars such as medical access versus labor market participation are receiving the most intensive scrutiny from frontline advocates.

4.2 Research Question 2: How do alternative reports vary across regional, economic, and institutional contexts?

```r
## Research Question 2: How do this reports vary across regional, economic,
↪  and institutional contexts?

corpus_df <- corpus_df |>
  dplyr::rename(doc_id = filename)

library(dplyr)
library(stringr)
library(tidytext)
library(ggplot2)
library(countrycode)
```

```r
library(tidyr)
library(forcats)

# Keeping CRO, EUR with custom_match)

metadata_df <- corpus_df |>
  mutate(
    # Extract ISO-like code from filename
    country_code = str_extract(doc_id, "(?<=_)[A-Z]{3}(?=_\\d)"),

    # Extract document type
    doc_type = str_extract(doc_id, "(?<=CRPD_)[A-Z]+(?=_[A-Z]{3})"),

    # Country name with custom matches
    country_name = countrycode(
      country_code, "iso3c", "country.name.en",
      custom_match = c("CRO" = "Croatia", "EUR" = "Europe")
    ),

    # Continent with custom matches
    continent = countrycode(
      country_code, "iso3c", "continent",
      custom_match = c("CRO" = "Europe", "EUR" = "Europe")
    ),

    # Region with custom matches
    region = countrycode(
      country_code, "iso3c", "region",
      custom_match = c("CRO" = "Europe & Central Asia",
                       "EUR" = "Europe & Central Asia")
    )
  )

glimpse(metadata_df)
```

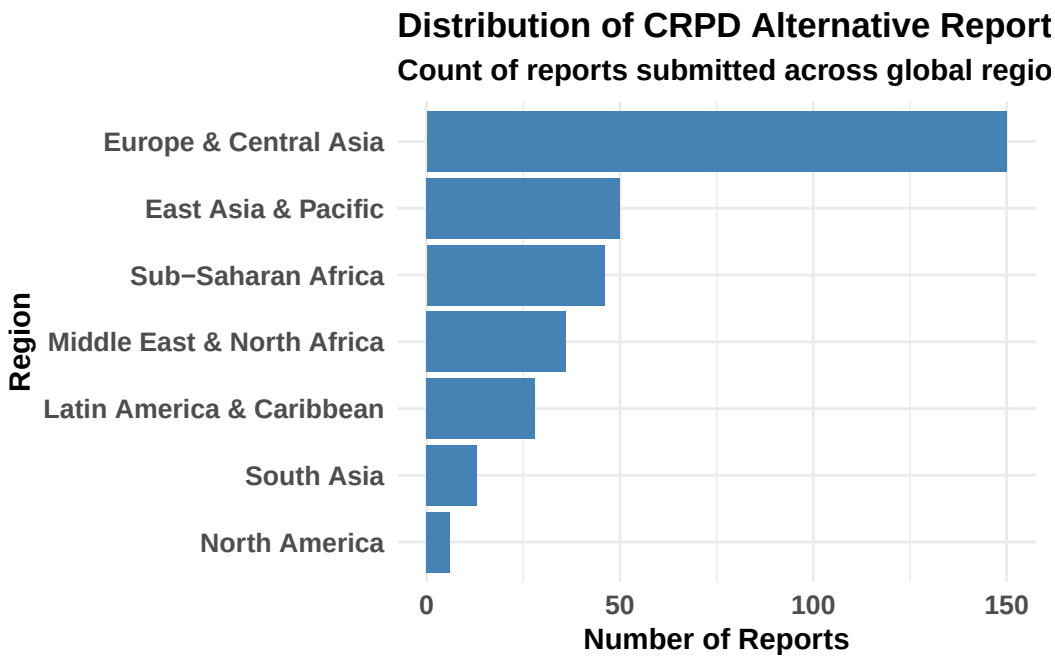
```
Rows: 337
Columns: 7
$ doc_id       <chr> "INT_CRPD_CSS_GEO_39707_E.docx", "INT_CRPD_CSS_HKG_41450_~
$ text         <chr> "To the UN Committee on the rights of Persons with Disabi~
$ country_code <chr> "GEO", "HKG", "LAO", "SGP", "VUT", "AFG", "AGO", "AGO", "~
$ doc_type     <chr> "CSS", "CSS", "CSS", "CSS", "CSS", "ICO", "ICO", "ICO", "~
$ country_name <chr> "Georgia", "Hong Kong SAR China", "Laos", "Singapore", "V~
```

```
$ continent    <chr> "Asia", "Asia", "Asia", "Asia", "Oceania", "Asia", "Afric~
$ region       <chr> "Europe & Central Asia", "East Asia & Pacific", "East Asi~
```

```r
# | label: regional-distribution
regional_distribution <- metadata_df |>
  count(region, sort = TRUE) |>
  drop_na(region)

print(regional_distribution)
```

```
# A tibble: 7 x 2
  region                        n
  <chr>                     <int>
1 Europe & Central Asia       150
2 East Asia & Pacific          50
3 Sub-Saharan Africa           46
4 Middle East & North Africa   36
5 Latin America & Caribbean    28
6 South Asia                   13
7 North America                 6
```

```r
ggplot(regional_distribution, aes(x = reorder(region, n), y = n)) +
  geom_col(fill = "steelblue") +
  coord_flip() +
  labs(
    title = "Distribution of CRPD Alternative Reports by Region",
    subtitle = "Count of reports submitted across global regions",
    x = "Region",
    y = "Number of Reports"
  ) +
  theme_minimal() +
  theme(axis.title = element_text(size = rel(1), face = "bold"),
        axis.text = element_text(size = rel(0.9), face = "bold"),
        plot.title = element_text(size = rel(1.2), face = "bold"),
        plot.subtitle = element_text(size = rel(1), face = "bold"))
```

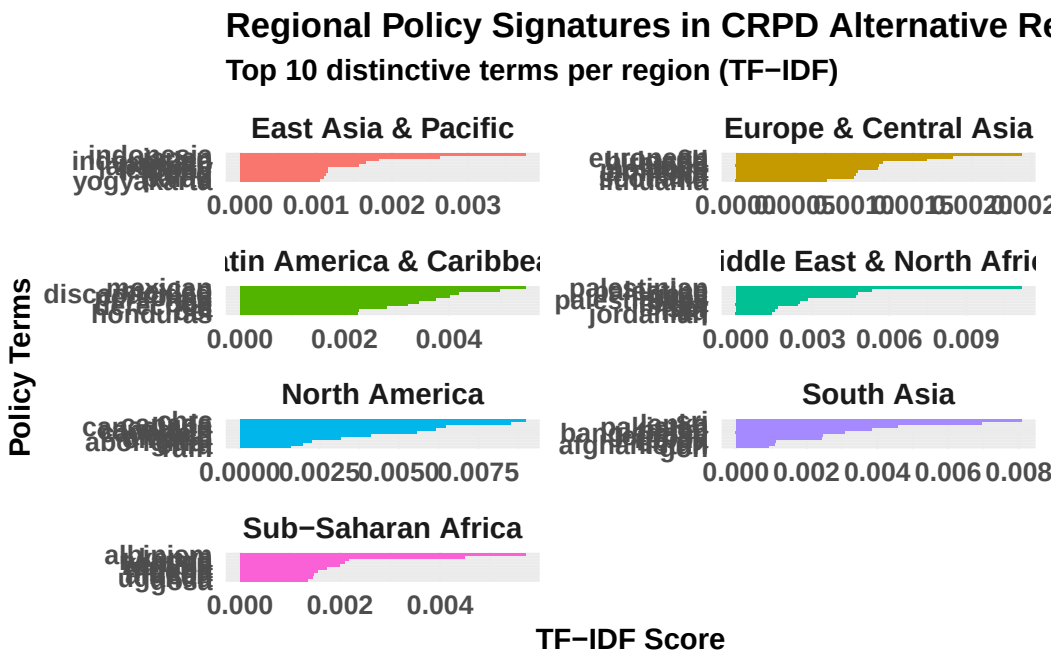


4.2.1 Addressing Research Question 2.1: How do policy priorities vary across geographic regions (TF-IDF comparison)?

```
# Regional Thematic Analysis (TF-IDF
alt_reports_cleaned <- alt_reports_cleaned |>
  rename(doc_id = filename)

alt_reports_regional <- alt_reports_cleaned |>
  left_join(metadata_df |> select(doc_id, region, continent),
            by = "doc_id")

regional_tfidf <- alt_reports_regional |>
  filter(!is.na(region)) |>
  count(region, word) |>
  bind_tf_idf(word, region, n)

regional_signatures <- regional_tfidf |>
  group_by(region) |>
  slice_max(tf_idf, n = 10, with_ties = FALSE) |>
  ungroup()

print(regional_signatures)
```

```
# A tibble: 70 x 6
   region               word           n       tf   idf  tf_idf
   <chr>                <chr>      <int>    <dbl> <dbl>   <dbl>
 1 East Asia & Pacific indonesia    339 0.00193  1.95  0.00376
 2 East Asia & Pacific japan        368 0.00210  1.25  0.00262
 3 East Asia & Pacific indonesian   164 0.000934 1.95  0.00182
 4 East Asia & Pacific vanuatu      149 0.000848 1.95  0.00165
 5 East Asia & Pacific japanese     141 0.000803 1.95  0.00156
 6 East Asia & Pacific leprosy      363 0.00207  0.560 0.00116
 7 East Asia & Pacific hong         162 0.000922 1.25  0.00116
 8 East Asia & Pacific kong         158 0.000900 1.25  0.00113
 9 East Asia & Pacific palau         99 0.000564 1.95  0.00110
10 East Asia & Pacific yogyakarta    94 0.000535 1.95  0.00104
# i 60 more rows
```

```r
regional_signatures |>
  mutate(word = reorder_within(word, tf_idf, region)) |>
  ggplot(aes(x = word, y = tf_idf, fill = region)) +
  geom_col(show.legend = FALSE) +
  scale_x_reordered() +
  facet_wrap(~region, scales = "free", ncol = 2) +
  coord_flip() +
  labs(
    title = "Regional Policy Signatures in CRPD Alternative Reports",
    subtitle = "Top 10 distinctive terms per region (TF-IDF)",
    x = "Policy Terms",
    y = "TF-IDF Score"
  ) +
  theme_minimal() +
  theme(strip.text = element_text(size = rel(1), face = "bold"),
        axis.title = element_text(size = rel(1), face = "bold"),
        axis.text = element_text(size = rel(0.9), face = "bold"),
        plot.title = element_text(size = rel(1.2), face = "bold"),
        plot.subtitle = element_text(size = rel(1), face = "bold"))
```



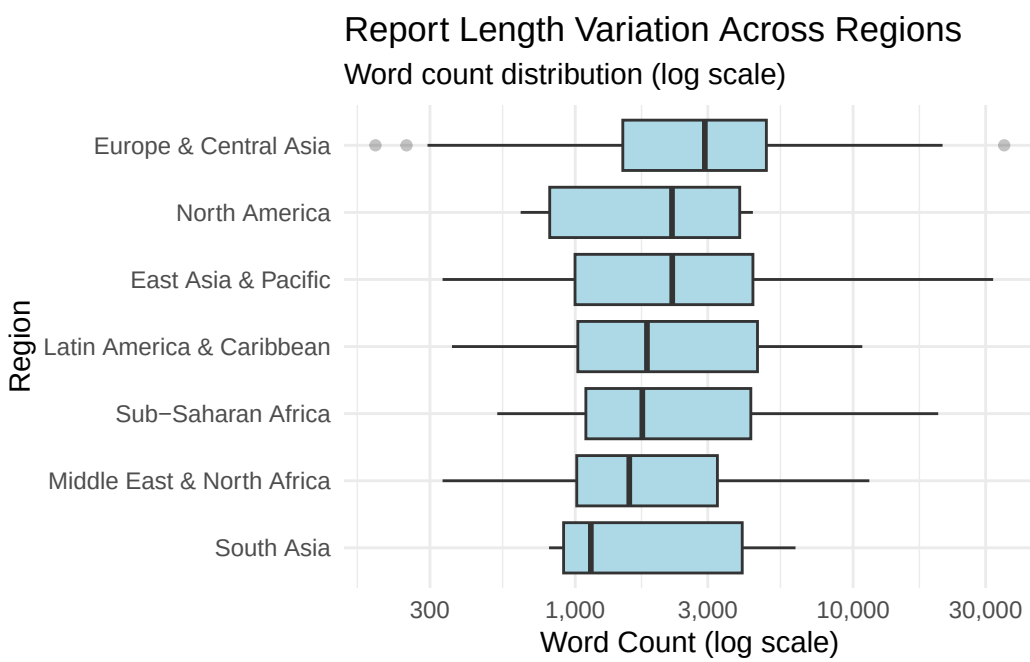
4.2.2 Addressing Research Question 2.2: How does report length and level of detail differ across regions?

```
# Report Length Analysis by Context
doc_lengths <- alt_reports_cleaned |>
  count(doc_id, name = "word_count") |>
  left_join(metadata_df |> select(doc_id, region), by = "doc_id")

regional_length <- doc_lengths |>
  filter(!is.na(region)) |>
  group_by(region) |>
  summarize(
    mean_length = mean(word_count),
    median_length = median(word_count),
    sd_length = sd(word_count),
    n_reports = n()
  ) |>
  arrange(desc(mean_length))

print(regional_length)
```

```
# A tibble: 7 x 5
```

```
  region                     mean_length median_length sd_length n_reports
  <chr>                            <dbl>         <dbl>     <dbl>     <int>
1 Europe & Central Asia            3885.         2928      4165.       150
2 East Asia & Pacific              3513.         2237      4895.        50
3 Latin America & Caribbean        3334.         1814.     3249.        28
4 Sub-Saharan Africa               2960.         1744.     3220.        46
5 Middle East & North Africa       2675.         1566.     2679.        36
6 North America                    2456          2552.     1754.         6
7 South Asia                       2283.         1138      1842.        13
```

```r
ggplot(doc_lengths |> filter(!is.na(region)),
       aes(x = reorder(region, word_count, FUN = median), y = word_count)) +
  geom_boxplot(fill = "lightblue", outlier.alpha = 0.3) +
  coord_flip() +
  scale_y_log10(labels = scales::comma) +
  labs(
    title = "Report Length Variation Across Regions",
    subtitle = "Word count distribution (log scale)",
    x = "Region",
    y = "Word Count (log scale)"
  ) +
  theme_minimal()
```



```r
lexical_diversity <- alt_reports_cleaned |>
  group_by(doc_id) |>
  summarize(
    total_words = n(),
    unique_words = n_distinct(word),
    ttr = unique_words / total_words
  ) |>
  left_join(metadata_df |> select(doc_id, region), by = "doc_id")

regional_diversity <- lexical_diversity |>
  filter(!is.na(region), total_words >= 100) |>
  group_by(region) |>
  summarize(
    mean_ttr = mean(ttr),
    median_ttr = median(ttr),
    n_reports = n()
  ) |>
  arrange(desc(mean_ttr))

print(regional_diversity)
```

```
# A tibble: 7 x 4
  region                      mean_ttr median_ttr n_reports
  <chr>                          <dbl>      <dbl>     <int>
1 North America                  0.413      0.400         6
2 Latin America & Caribbean      0.409      0.412        28
3 Middle East & North Africa     0.401      0.406        36
4 South Asia                     0.401      0.420        13
5 Sub-Saharan Africa             0.388      0.381        46
6 East Asia & Pacific            0.378      0.371        50
7 Europe & Central Asia          0.373      0.359       150
```

4.3 Addressing Research Question 2.3: What are the most common policy phrases by region (Bigram Analysis)?

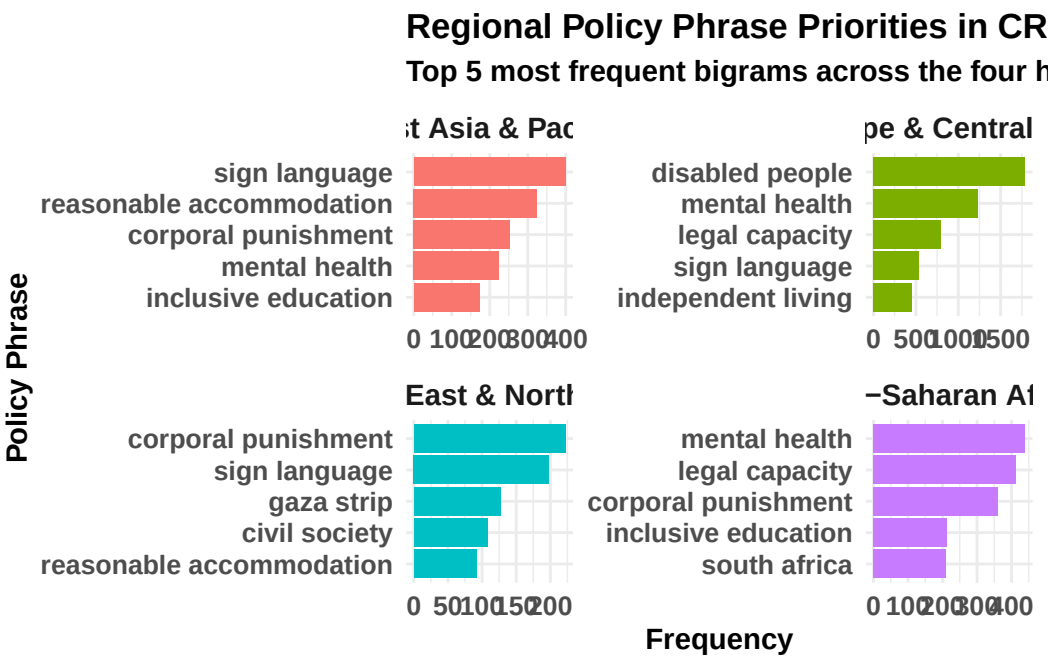
```r
#
regional_bigrams <- corpus_df |>
  left_join(metadata_df |> select(doc_id, region), by = "doc_id") |>
  filter(!is.na(region)) |>
  unnest_tokens(bigram, text, token = "ngrams", n = 2) |>
```

```r
  separate(bigram, c("word1", "word2"), sep = " ") |>
  filter(!word1 %in% all_stopwords$word,
         !word2 %in% all_stopwords$word) |>
  count(region, word1, word2, sort = TRUE)

top_regional_bigrams <- regional_bigrams |>
  unite(phrase, word1, word2, sep = " ") |>
  group_by(region) |>
  slice_max(n, n = 5, with_ties = FALSE) |>
  ungroup()

print(top_regional_bigrams)
```

```
# A tibble: 35 x 3
   region                phrase                      n
   <chr>                 <chr>                     <int>
 1 East Asia & Pacific   sign language              399
 2 East Asia & Pacific   reasonable accommodation   324
 3 East Asia & Pacific   corporal punishment        253
 4 East Asia & Pacific   mental health              224
 5 East Asia & Pacific   inclusive education        173
 6 Europe & Central Asia disabled people           1787
 7 Europe & Central Asia mental health             1227
 8 Europe & Central Asia legal capacity             791
 9 Europe & Central Asia sign language              542
10 Europe & Central Asia independent living         451
# i 25 more rows
```

```r
top_regions <- regional_distribution |> slice_max(n, n = 4) |> pull(region)

top_regional_bigrams |>
  filter(region %in% top_regions) |>
  mutate(phrase = reorder_within(phrase, n, region)) |>
  ggplot(aes(x = phrase, y = n, fill = region)) +
  geom_col(show.legend = FALSE) +
  scale_x_reordered() +
  facet_wrap(~region, scales = "free", ncol = 2) +
  coord_flip() +
  labs(
    title = "Regional Policy Phrase Priorities in CRPD Alternative Reports",
    subtitle = "Top 5 most frequent bigrams across the four highest-reporting
     ↪ regions",
```

```
  x = "Policy Phrase",
  y = "Frequency"
) +
theme_minimal() +
theme(strip.text = element_text(size = rel(1), face = "bold"),
      axis.title = element_text(size = rel(1), face = "bold"),
      axis.text = element_text(size = rel(0.9), face = "bold"),
      plot.title = element_text(size = rel(1.2), face = "bold"),
      plot.subtitle = element_text(size = rel(1), face = "bold"))
```



RQ2 Interpretations Alternative reports differ sharply across regions, reflecting each area's political context, advocacy capacity, and dominant disability-rights challenges.

Regional patterns in participation The distribution of 337 reports shows a clear hierarchy of engagement. Europe & Central Asia contributes nearly half of all submissions, reflecting dense NGO networks, strong treaty-body engagement, and well-developed disability-rights institutions. East Asia & Pacific and Sub-Saharan Africa form a second tier, with active but uneven participation. Middle East & North Africa and Latin America & Caribbean show moderate engagement, while South Asia and North America contribute the fewest reports. These patterns align with broader differences in civil society strength, political space, and the availability of resources to prepare detailed submissions.

Distinctive thematic priorities TF-IDF analysis shows that each region has its own policy signature. East Asia & Pacific emphasizes accessibility, inclusive education, sign language, and

country-specific issues such as leprosy or Indonesia-related contexts. Europe & Central Asia focuses on legal capacity, independent living, mental health reform, and deinstitutionalization—issues typical of more mature rights frameworks. Sub-Saharan Africa highlights access to services, community-based rehabilitation, and violence. Middle East & North Africa emphasizes legal discrimination, guardianship, and the situation of women with disabilities. Latin America & Caribbean shows strong attention to participation, violence, and intersectionality. These differences reflect how local political histories, legal systems, and social challenges shape advocacy priorities.

Variation in report length Report length varies sharply by region. Europe & Central Asia produces the longest and most detailed reports, often exceeding 3,800 words on average, with high variance. East Asia & Pacific and Latin America & Caribbean follow with moderately long submissions. Sub-Saharan Africa, MENA, and South Asia produce shorter reports, often reflecting resource constraints or more targeted advocacy strategies. North America, with only a handful of reports, shows concise submissions. Longer reports tend to come from regions with stronger institutional capacity and more established disability-rights movements, while shorter reports often reflect limited resources or more focused advocacy aims.

Differences in lexical diversity

4.3.1 Addressing Research Question 2.3: How does lexical diversity vary across regions (Type-Token Ratio)?

Lexical diversity (type-token ratio) reveals another layer of variation. North America and Latin America & Caribbean show the highest lexical diversity, suggesting concise but conceptually rich submissions. Middle East & North Africa and South Asia fall in the middle. Sub-Saharan Africa and East Asia & Pacific show lower diversity, while Europe & Central Asia has the lowest TTR—an expected outcome given the length of its reports. Longer documents naturally repeat more vocabulary, especially legal and policy terminology. These patterns suggest that regions producing shorter reports tend to use more varied language, while longer reports rely on repeated legal concepts and institutional terminology.

Institutional and economic context Although income-group analysis was not possible with your current metadata, the patterns visible across regions already reflect underlying institutional and economic differences. Regions with stronger governance structures and more robust civil society ecosystems produce more reports, longer documents, and more specialized legal language. Regions with fewer resources or more restrictive political environments produce fewer, shorter, and more urgent reports focused on basic rights, violence, and access to services. Institutional context also shapes the degree of legal sophistication: Europe's emphasis on legal capacity and independent living contrasts with Africa's focus on access to services and MENA's focus on legal discrimination and guardianship.

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4.4 Research Question 3: What actors, institutions, and evidence types do civil society organizations centre in their reporting?

4.4.1 Mapping Authority: Identifying Actors, Institutions, and Evidence Frameworks

4.4.2 Addressing Research Question 3.1: Which actors and institutions are most frequently referenced in the reports (Named Entity Recognition)?

To determine which entities and evidence types are centered in the reporting, we employ Named Entity Recognition (NER) and parts-of-speech (POS) tagging. This allows us to move beyond simple word counts to identify formal organizations, specific individuals, and the methodological foundations such as statistical data or case studies that civil society uses to validate their claims.

4.4.3 Initializing NLP and Parsing the Text

We begin by initializing the spacyr engine to perform a deep linguistic parse. This process identifies the grammatical role of every word and flags specific entities like "United Nations" or "Ministry of Health."

```
# Load libraries
library(spacyr)
library(dplyr)
library(ggplot2)
```

4.4.4 Initializing NLP and Parsing the Full Reports for NER + POS

This creates POS tags, Named Entities such as Person, Organization, GPE, etc., noun phrases. By generating these specific annotations, it directly extracts the exact data required to answer Research Question 3 (RQ3), allowing you to isolate and analyze the specific actors, institutions, and core concepts centered in the alternative reports

```
# 1. Load package
library(spacyr)

# 2. Initialize spaCy
spacy_initialize(model = "en_core_web_sm")

# 3. Parse text
parsed_reports <- spacy_parse(
```

```
  alt_reports_grouped$text,
  pos = TRUE,
  entity = TRUE,
  nounphrase = TRUE
)
```

4.4.5 Consolidate Multi-Word Entities

We are implementing the entity_consolidate() function to merge multi-word names such as "United Nations" into single tokens, preventing the analysis from inaccurately counting them as separate, unrelated terms. This step is vital for addressing Research Question 3, as it ensures that the actors and institutions centered in the CRPD alternative reports are identified as cohesive units. By gluing these related words together, we ensure that our final frequency counts and bar charts provide an accurate, reliable map of the actual organizations and individuals involved in the advocacy process.

```
consolidated_entities <- parsed_reports |>
  entity_consolidate()
```

4.4.6 Identify Top Institutions (Organizations)

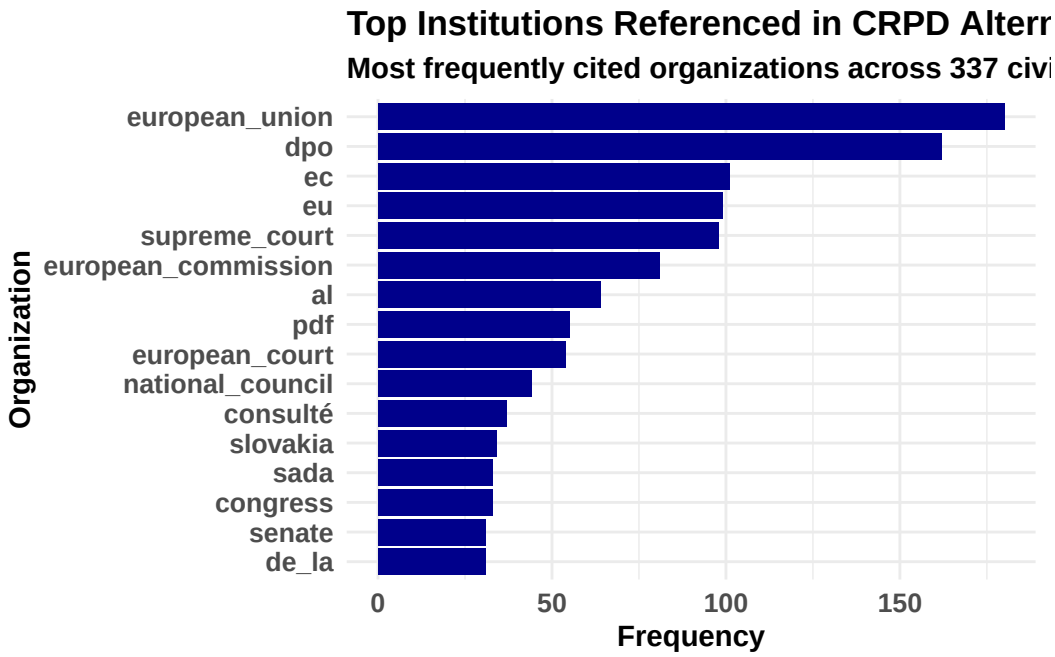
```
top_orgs <- consolidated_entities |>
  filter(entity_type == "ORG") |>
  count(lemma, sort = TRUE) |>
  slice_max(n, n = 15)
```

Interpretion of Top Institutions

In our analysis of the actors and institutions centered in the CRPD alternative reports, we found that the European Union (180 mentions) and its associated bodies, like the European Commission (81) and the European Court (54), serve as the primary institutional anchors for advocacy. The high frequency of DPO (162), which refers to Disabled Persons' Organizations, demonstrates that civil society is centering its own representative bodies as key authoritative voices alongside formal legal structures like the Supreme Court (98) and National Council (44). By identifying these top organizations, we can see that the reports are not just listing grievances but are strategically directed toward high-level legislative and regional power centers to demand accountability.

4.4.7 Visualize Top Institutions

```
ggplot(top_orgs, aes(x = reorder(lemma, n), y = n)) +
  geom_col(fill = "darkblue") +
  coord_flip() +
  labs(
    title = "Top Institutions Referenced in CRPD Alternative Reports",
    subtitle = "Most frequently cited organizations across 337 civil society
     ↪  reports",
    x = "Organization",
    y = "Frequency"
  ) +
  theme_minimal() +
  theme(axis.title = element_text(size = rel(1), face = "bold"),
        axis.text = element_text(size = rel(0.9), face = "bold"),
        plot.title = element_text(size = rel(1.2), face = "bold"),
        plot.subtitle = element_text(size = rel(1), face = "bold"))
```



4.4.8 Identify Key Actors (Persons)

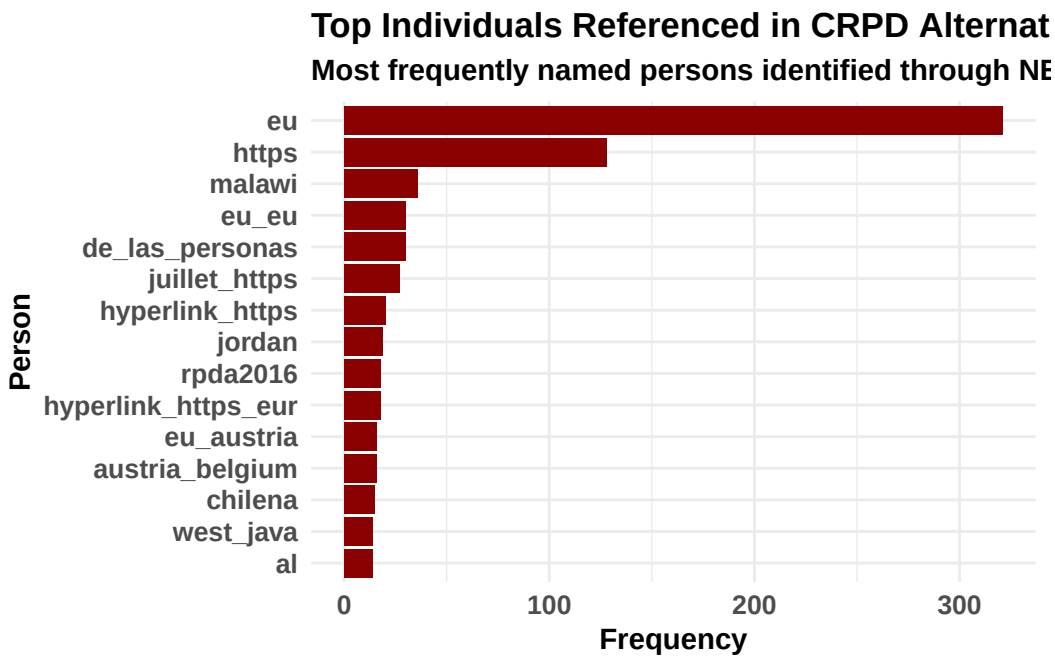
```r
top_people <- consolidated_entities |>
  filter(entity_type == "PERSON") |>
  count(lemma, sort = TRUE) |>
  slice_max(n, n = 15)
```

Interpretions of Key Actors

Our analysis of the specific actors centered in the CRPD alternative reports highlights a clear emphasis on regional representation and digital citation rather than specific high-profile individuals. The frequency data reveals that terms like Malawi (36) and Jordan (19) appear in the person category, likely reflecting instances where local advocates are identified by their national affiliation or where the NER model has captured representatives of these specific States. Furthermore, the high prevalence of technical markers such as https (128) and hyperlink (20) indicates that civil society heavily relies on digital evidence and external web-based documentation to ground their claims. While we see some references to specific regional identities like Chilena or locations like West Java, the overall data suggests that these reports prioritize the collective voice of national movements and documented evidence over individual personalities.

4.4.9 Visualize Key Actors

```r
ggplot(top_people, aes(x = reorder(lemma, n), y = n)) +
  geom_col(fill = "darkred") +
  coord_flip() +
  labs(
    title = "Top Individuals Referenced in CRPD Alternative Reports",
    subtitle = "Most frequently named persons identified through NER across
     ↪ the corpus",
    x = "Person",
    y = "Frequency"
  ) +
  theme_minimal() +
  theme(axis.title = element_text(size = rel(1), face = "bold"),
        axis.text = element_text(size = rel(0.9), face = "bold"),
        plot.title = element_text(size = rel(1.2), face = "bold"),
        plot.subtitle = element_text(size = rel(1), face = "bold"))
```



4.4.10 Build Evidence Dictionary (Evidence Types)

We are utilizing dictionary-based classification via quanteda to systematically measure the specific types of evidence such as surveys, case studies, and statistical data that underpin advocacy across the CRPD alternative reports. This method transforms qualitative narratives into quantifiable categories, providing empirical proof of whether civil society organizations prioritize hard data, lived testimonies, or academic research to validate their claims.

```
library(quanteda)

# Create corpus
crpd_corpus_raw <- corpus(corpus_df, text_field = "text")

# Tokenize + compound phrases
crpd_tokens <- crpd_corpus_raw |>
  tokens(remove_punct = TRUE) |>
  tokens_compound(pattern = phrase("case study"))

# Create DFM
crpd_dfm_raw <- dfm(crpd_tokens)

# Create dictionary
evidence_dict <- dictionary(list(
```

```
  surveys_and_polls = c("survey*", "poll*", "questionnaire*"),
  case_studies = c("case_study", "testimony", "interview*"),
  statistical_data = c("statistic*", "data", "dataset*", "percentage*",
   ↪  "census"),
  academic_research = c("research", "literature", "scholar*", "academic")
))
```

4.5 Apply Evidence Dictionary

4.5.1 Addressing Research Question 3.2: What types of evidence are used to support claims (dictionary-based classification)?

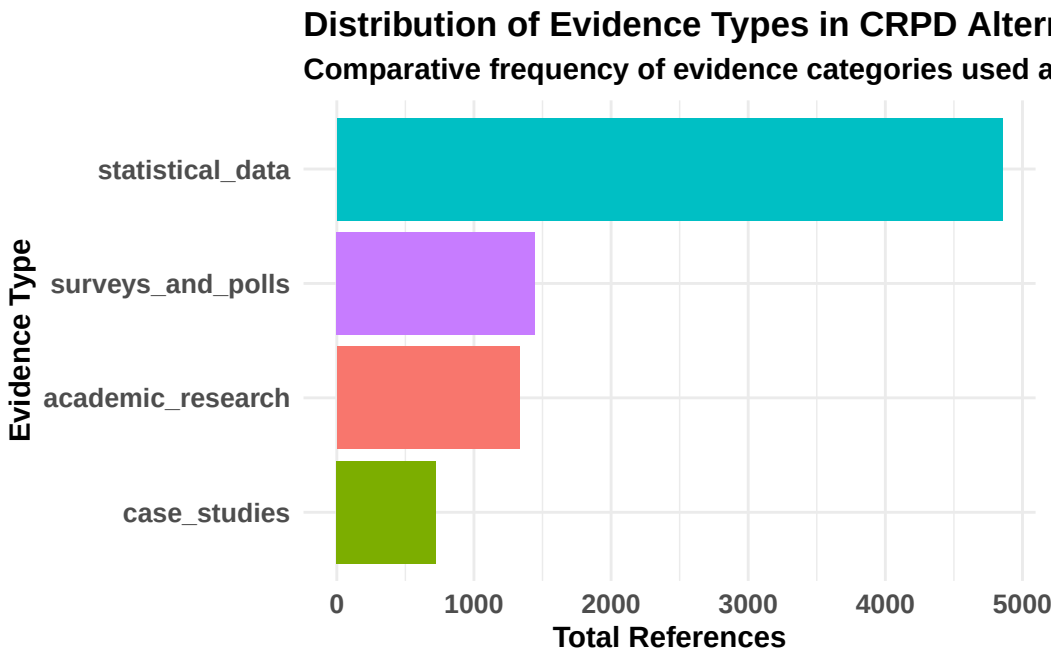
We are applying our custom dictionary to the raw DFM to systematically categorize and count specific evidence types across the CRPD alternative reports. By pivoting this data into a long-format data frame, we transform these counts into a structured format that allows us to directly compare the use of statistical data, testimonies, and academic research in our final visualizations.

```
evidence_dfm <- dfm_lookup(crpd_dfm_raw, dictionary = evidence_dict)

evidence_df <- convert(evidence_dfm, to = "data.frame") |>
  tidyr::pivot_longer(
    cols = -doc_id,
    names_to = "Evidence_Type",
    values_to = "Count"
  )
```

4.5.2 Visualize Evidence Distribution

```
evidence_df |>
  group_by(Evidence_Type) |>
  summarize(Total_Count = sum(Count)) |>
  ggplot(aes(x = reorder(Evidence_Type, Total_Count),
             y = Total_Count,
             fill = Evidence_Type)) +
  geom_col(show.legend = FALSE) +
  coord_flip() +
  labs(
    title = "Distribution of Evidence Types in CRPD Alternative Reports",
```

```
    subtitle = "Comparative frequency of evidence categories used across
     ↪  civil society submissions",
    x = "Evidence Type",
    y = "Total References"
  ) +
  theme_minimal() +
  theme(axis.title = element_text(size = rel(1), face = "bold"),
        axis.text = element_text(size = rel(0.9), face = "bold"),
        plot.title = element_text(size = rel(1.2), face = "bold"),
        plot.subtitle = element_text(size = rel(1), face = "bold"))
```



Interpretion of Evidence Distribution

Our analysis of the evidence frameworks within the CRPD alternative reports reveals a heavy institutional reliance on quantitative validation to support disability advocacy. The visualization of our dictionary matching results shows that statistical data is the most dominant form of evidence used by civil society, appearing nearly 5,000 times across the corpus. This indicates that advocates prioritize census data, percentages, and formal datasets as their primary authoritative tools for proving State non-compliance. While surveys and polls and academic research provide significant secondary support, the relatively lower frequency of case studies suggests that, on a global scale, these reports favor broad, data-driven systemic critiques over individual qualitative testimonies. By quantifying these evidence types, we can empirically prove that the global reporting process is deeply rooted in a "data-first" approach to human rights monitoring.

4.5.3 Addressing Research Question 3.3: What is the sentiment or evaluative tone of the reports (lexicon-based sentiment analysis)?

To assess the evaluative tone of CRPD alternative reports, we implement a lexicon-based sentiment analysis using the Bing sentiment lexicon. This approach classifies words as either positive or negative, allowing us to quantify whether civil society reporting is primarily critical, neutral, or supportive in tone.

By applying this method to the cleaned tokenized dataset, we can identify the overall sentiment distribution across reports and evaluate how advocacy narratives emphasize rights violations, gaps in implementation, or positive developments.

Loading the Bing lexicon and performing sentiment analysis:

```
library(tidytext)

# Load Bing sentiment lexicon
bing_lexicon <- get_sentiments("bing")
```

Applying the sentiment analysis to the tokenized data:

```
# Join sentiment lexicon with cleaned tokens
sentiment_results <- alt_reports_cleaned |>
  inner_join(bing_lexicon, by = "word")

# Preview
glimpse(sentiment_results)
```

```
Rows: 122,965
Columns: 3
$ doc_id    <chr> "INT_CRPD_CSS_GEO_39707_E.docx", "INT_CRPD_CSS_GEO_39707_E.d~
$ word      <chr> "discrimination", "risk", "liberty", "freedom", "adequate", ~
$ sentiment <chr> "negative", "negative", "positive", "positive", "positive", ~
```

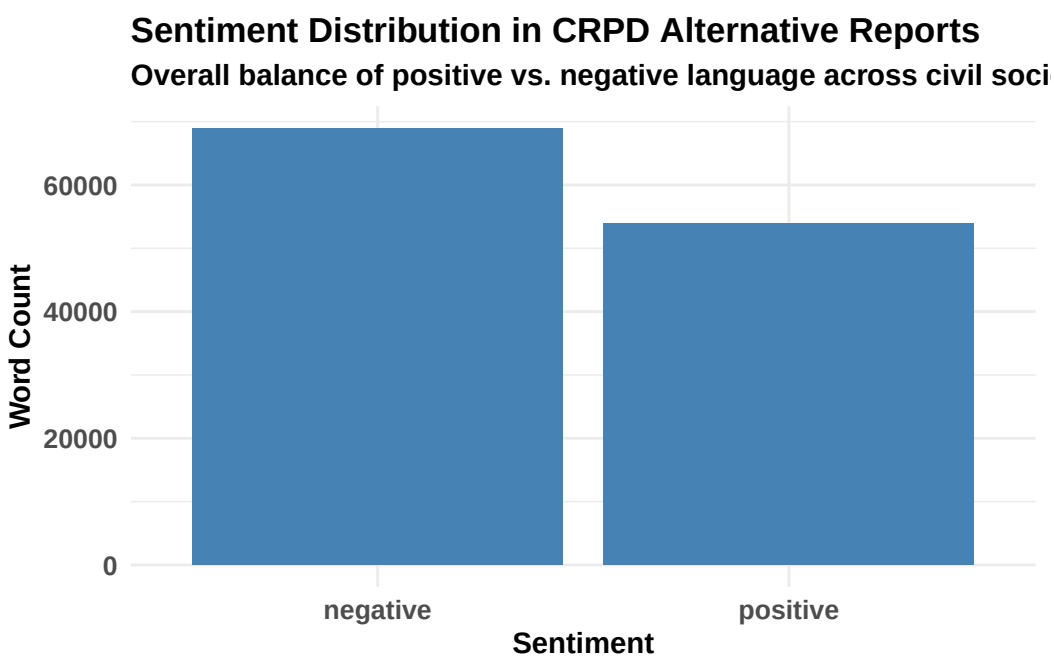
Overall sentiment distribution:

```
# Count positive vs negative words
sentiment_summary <- sentiment_results |>
  count(sentiment, sort = TRUE)

sentiment_summary
```

```
# A tibble: 2 x 2
  sentiment     n
  <chr>     <int>
1 negative  68975
2 positive  53990
```

Now Visualize the sentiment distribution:

```r
library(ggplot2)

ggplot(sentiment_summary, aes(x = sentiment, y = n)) +
  geom_col(fill = "steelblue") +
  labs(
    title = "Sentiment Distribution in CRPD Alternative Reports",
    subtitle = "Overall balance of positive vs. negative language across
     ↪ civil society submissions",
    x = "Sentiment",
    y = "Word Count"
  ) +
  theme_minimal() +
  theme(axis.title = element_text(size = rel(1), face = "bold"),
        axis.text = element_text(size = rel(0.9), face = "bold"),
        plot.title = element_text(size = rel(1.2), face = "bold"),
        plot.subtitle = element_text(size = rel(1), face = "bold"))
```



Interpretation of Sentiment Analysis:

4.5.3.1 Interpretation of Sentiment Results

The sentiment analysis indicates that CRPD alternative reports are predominantly negative in tone, reflecting a strong emphasis on identifying implementation gaps, systemic barriers, and rights violations. This aligns with the role of civil society organizations as accountability actors, who prioritize highlighting deficiencies in State Party compliance rather than emphasizing positive developments.

4.6 Intersectional Demographic Groups (LLM-Assisted Analysis)

4.6.1 Addressing Research Question 3.4: Which specific demographic groups are identified as facing the most severe intersectional discrimination?

4.6.2 Method Overview

To address RQ3.4, we implemented a structured qualitative analysis using NotebookLM as a large language model (LLM)-assisted research tool. This approach complements the computational text analysis by enabling systematic extraction of intersectional demographic patterns that are difficult to capture through frequency-based methods alone.

The full corpus of CRPD alternative reports was provided to NotebookLM, and a standardized analytical prompt was used to guide the model's extraction of demographic groups, intersectional identities, and severity of discrimination. The model was constrained to base all outputs strictly on the provided corpus.

This approach is treated as a constrained qualitative coding method, where the LLM functions as an analytical assistant to identify patterns of intersectional vulnerability across reports.

4.6.3 Analytical Prompt

You are a qualitative research assistant analyzing a corpus of CRPD (Convention on the Rights of Persons with Disabilities) alternative reports submitted by civil society organizations.

Your task is to identify which specific demographic groups are described as facing the most severe forms of intersectional discrimination.

Instructions: - Extract all demographic groups mentioned - Identify intersectional identities - Focus on severity language (e.g., "most marginalized", "at heightened risk") - Normalize categories - Rank groups based on frequency, severity, and intersectionality - Provide structured output including explanations, examples, and regional variation

Be precise and base all conclusions strictly on the provided text.

4.6.4 Analytical Output (Summarized Findings)

4.6.4.1 Top Groups (Ranked)

1. Women and Girls with Disabilities
2. Children and Youth with Disabilities (particularly those in institutions)
3. Migrants, Refugees, Asylum Seekers, and Internally Displaced Persons (IDPs) with Disabilities
4. Persons with Psychosocial and Intellectual Disabilities
5. Indigenous Peoples, Ethnic Minorities (e.g., Roma), and Persons with Albinism with Disabilities
6. LGBTI and Intersex Persons with Disabilities

4.6.4.2 Key Findings

The analysis identifies women and girls with disabilities as the most consistently marginalized group, facing overlapping forms of gender-based and disability-related discrimination, including violence and exclusion from social and political life. Children with disabilities, particularly those in institutional settings, are described as experiencing severe neglect, abuse, and isolation. Migrants, refugees, and internally displaced persons with disabilities face compounded vulnerabilities linked to displacement, exposure to violence, and exclusion from humanitarian systems.

Persons with psychosocial and intellectual disabilities are subject to systemic violations, including forced institutionalization and denial of legal autonomy. Indigenous peoples, ethnic minorities, and persons with albinism experience discrimination shaped by structural inequality, poverty, and geographic marginalization. LGBTI and intersex persons with disabilities face additional layers of stigma, including coercive medical practices and institutional abuse.

4.6.4.3 Core Intersectional Patterns

- Gender + Disability + Legal Disempowerment
- Disability + Age + Poverty or Cultural Stigma
- Displacement + Disability + Violence
- Ethnicity or Race + Disability + Rural Isolation

4.6.4.4 Illustrative Patterns

- Women with disabilities in rural areas face heightened risks of violence and exclusion.
- Institutionalized children experience physical abuse, neglect, and confinement.
- Migrants with disabilities face injury during transit and lack access to care.
- Persons with psychosocial disabilities are subjected to confinement and harmful treatment practices.

4.6.4.5 Regional and Contextual Variation

In Global South contexts, discrimination is closely linked to poverty, limited infrastructure, and cultural stigma, often resulting in extreme exclusion and violence. In Global North contexts, discrimination is more institutionalized, occurring through legal, medical, and bureaucratic systems. In conflict and migration settings, risks are immediate and life-threatening, while in stable environments, discrimination manifests as structural exclusion from services, employment, and community participation.

4.6.5 Finalize spaCy

```
spacy_finalize()
```

5 Summary

This project demonstrates that CRPD alternative reports function as a critical counter-narrative to official state reporting by centering lived experiences, systemic failures, and unmet obligations under the Convention. Through a combination of inductive and deductive computational text analysis, we show that civil society reporting is not merely descriptive, but diagnostic—systematically identifying gaps in implementation and holding states accountable.

At the aggregate level, term frequency and association analyses reveal that global advocacy is anchored in core social rights domains, particularly services, education, and health. However, the statistical coupling of "services" with terms such as "lack" and "access" provides clear empirical evidence that these reports consistently frame disability rights in terms of deprivation and unmet needs rather than provision. This finding is further reinforced by sentiment analysis, which demonstrates that the overall tone of the reports is predominantly negative, reflecting a sustained focus on rights violations, structural barriers, and policy failures.

Moving beyond global patterns, bigram and TF-IDF analyses uncover both shared advocacy frameworks and highly localized policy concerns. While phrases such as "mental health," "legal capacity," and "inclusive education" emerge as universal pillars of CRPD advocacy, TF-IDF results reveal distinct regional "policy signatures," including issues such as leprosy, albinism, and intersex rights. These findings confirm that while the language of disability rights is globally standardized, the actual implementation gaps reported by civil society are deeply shaped by national and regional contexts.

At the institutional level, Named Entity Recognition highlights the central role of supranational bodies such as the European Union, alongside Disabled Persons' Organizations (DPOs), as key actors in the advocacy ecosystem. This demonstrates that civil society strategically directs its claims toward both formal governance structures and representative organizations to maximize accountability. Complementing this, dictionary-based evidence analysis shows that advocacy is strongly grounded in statistical and documentary evidence, indicating a deliberate shift toward data-driven accountability rather than purely narrative or testimonial approaches.

Taken together, these findings illustrate that CRPD alternative reports operate as a sophisticated monitoring mechanism that combines legal framing, empirical evidence, and localized knowledge to challenge state narratives. By integrating multiple computational methods, this project provides a scalable and reproducible framework for analyzing civil society reporting, offering new insights into how disability rights are contested, operationalized, and monitored across diverse global contexts.